Manual

Advanced Configurations: MES Terminal AIP

EAT-AIP 8.1

Version 1.4.5987

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Copyright

Contents

1 Extended Application Training: MES Terminal

Purpose

The configuration options described in this function package enable varied modifications to dialogs, dialog fields, formats, buttons of the HYDRA shop floor program AIP (Acquisition and Information Panel).

Implementation notes

The function package is used if you would like to change

- the dialog structure
- dialog fields, labeling and units
- data types of dialogs including value ranges
- buttons and labeling
- the presentation of columns and if you would like to add further columns to lists

Integration

The AIP terminal provides various options to change dialogs. Changes are either carried out directly at the shop floor client or in the dynamic dialog configuration of the MOC if the presentation of input dialogs should be changed.

Features

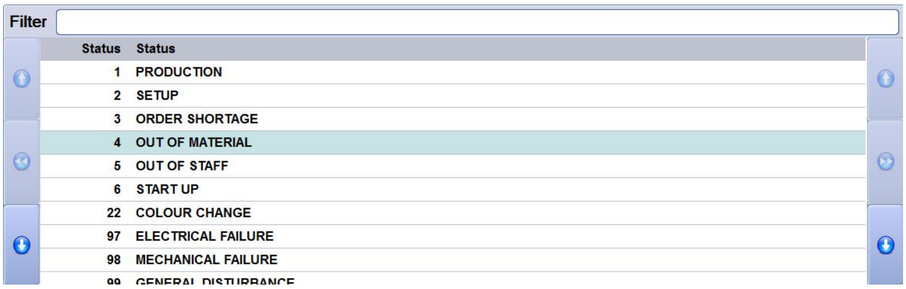
- General terminal configurations
- Configuration of grid layout
- Button configuration (basic screen)
- Configurations for the virtual keyboard
- Configurations for barcode input
- Dynamic dialog configuration for input fields and dialog buttons

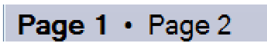
2 Operation of AIP

2.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, …) as well as the selection lists of posting dialogs.

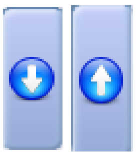


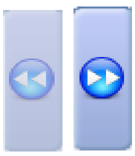


Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys:'↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.

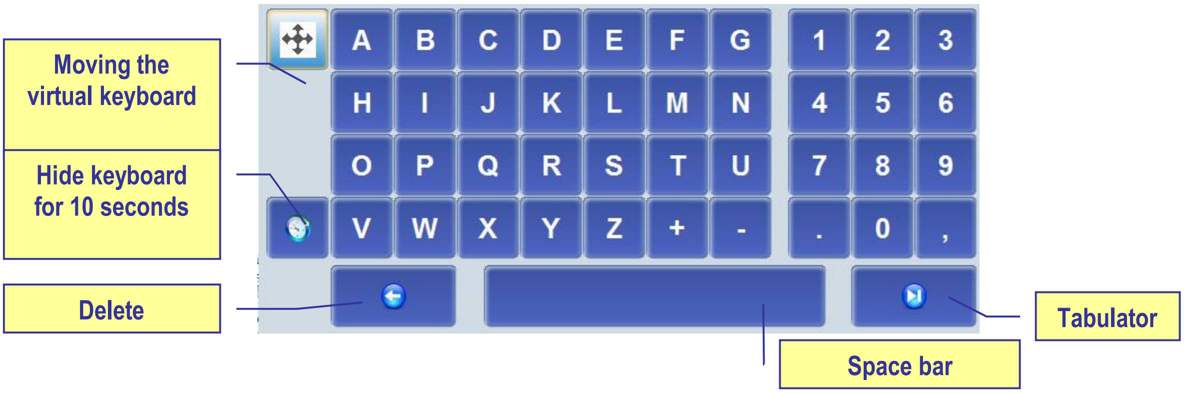

Scrolling page by page (up or down) in the table.


Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

A “table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.



The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+“, "-“ , ".“ and ",“. Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter –t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-“, “/“ and “.“ are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- English(USA) MM/DD/YYYY
- Danish MM-DD-YYYY
- Customer-specific 1 YYYY-MM-DD
- Customer-specific 2 YYYY/MM/DD
- Customer-specific 3 MM/YYYY/DD

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

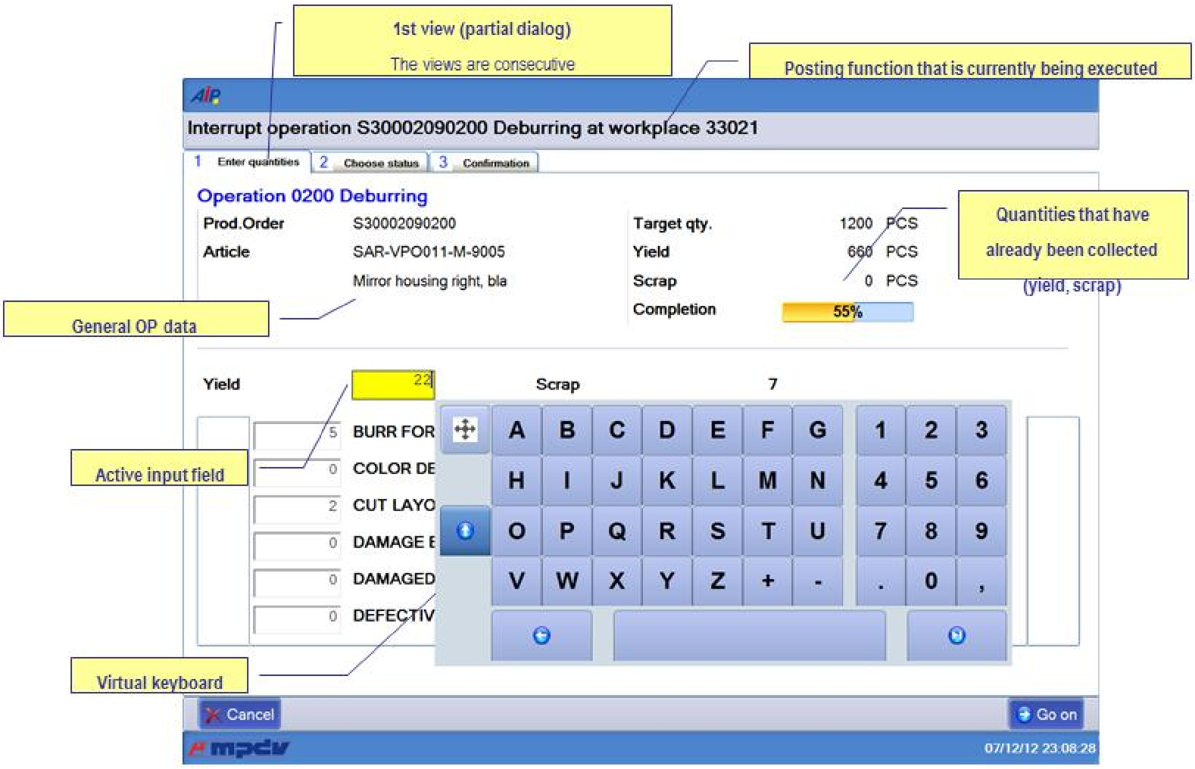
The year is displayed only by two characters in the status bar.

2.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:

The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.



The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

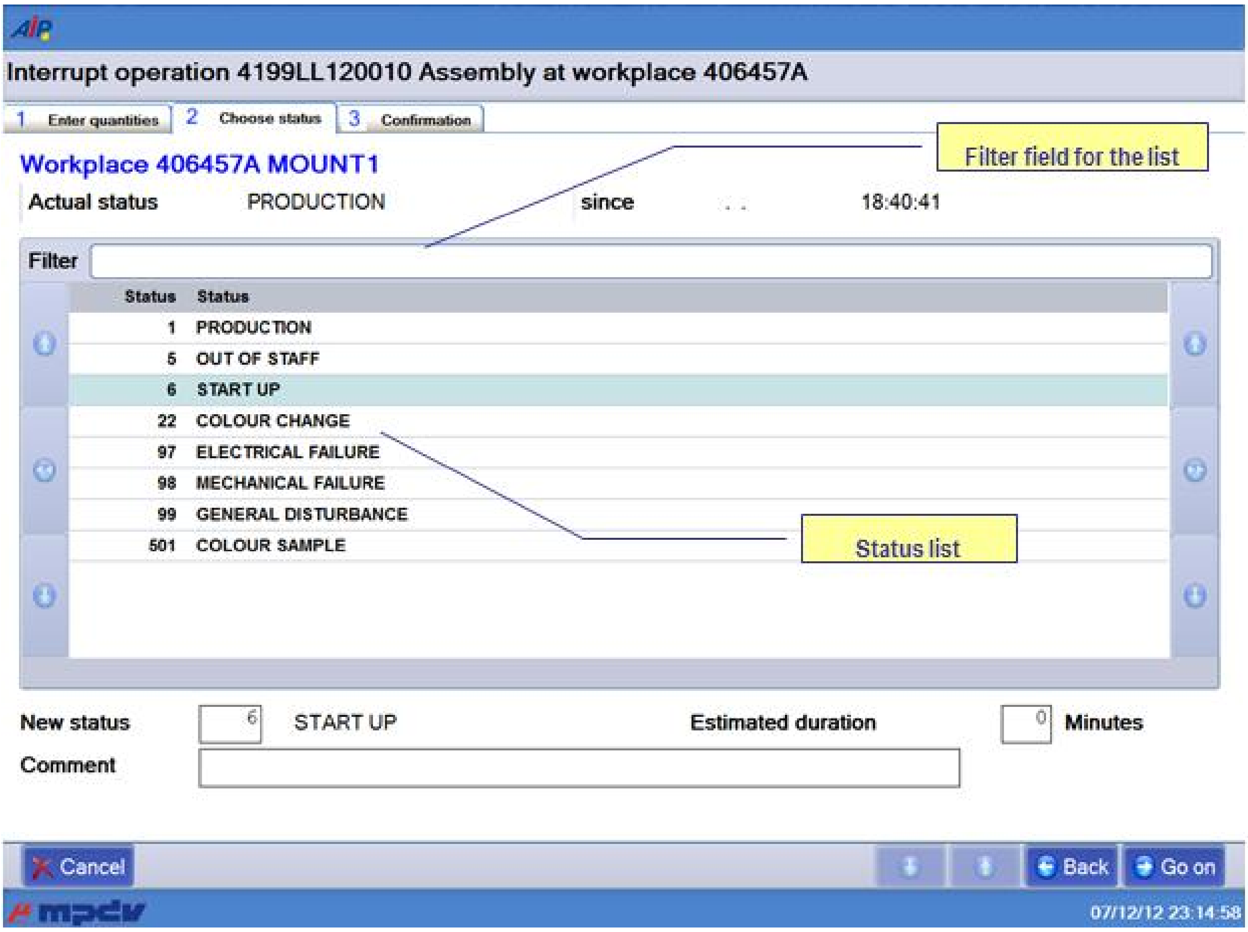
Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

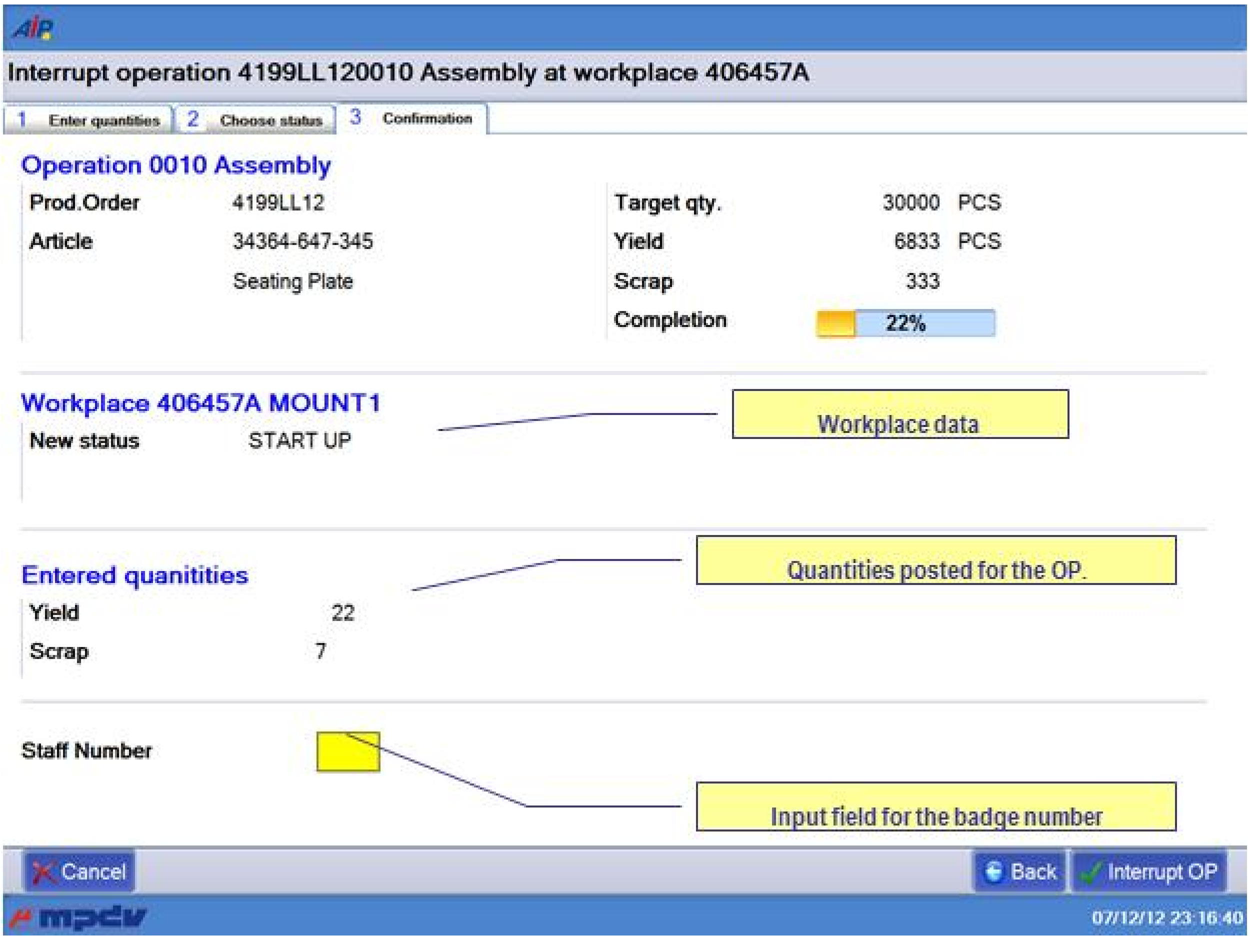
The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.



The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).



The partial dialog "confirm" shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the "interrupt operation" dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.

If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.

3 Basic AIP Display

	→	

In general, *AIP* has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons within the touch screen or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these selection lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.

To ensure proper processing and posting, terminals with "MDE" operation mode must <u>not</u> be switched off during times without shift.

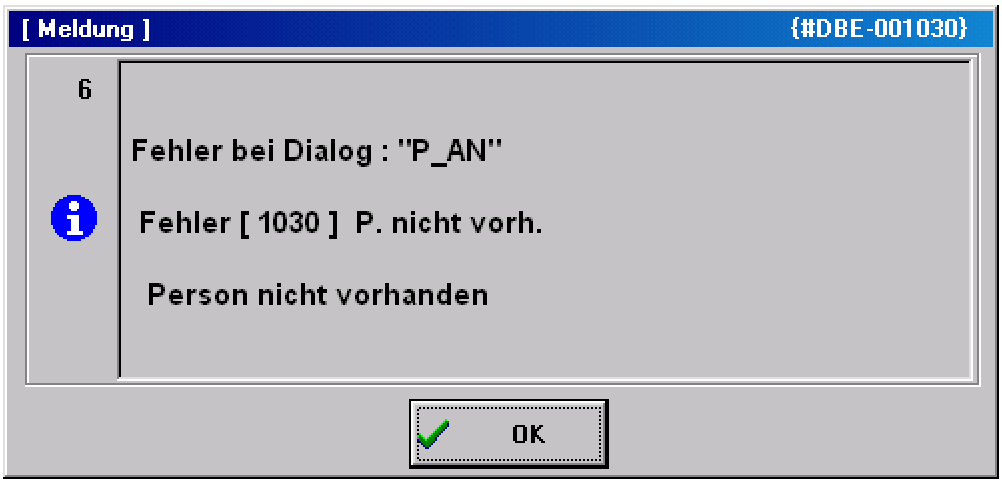
3.1 Basic displays – header and footer

Header

The *AIP* logo is displayed top left of the screen, which may be exchanged by a customer logo after corresponding configuration.

Possible messages (e.g. if a dialog is opened for more than five minutes) are displayed to the right of it.

A separate window opens to display error messages that occur during data collection (e.g. validity checks).



Basic displays

A maximum of 16 workplaces or machines can be assigned to the *AIP* terminal. The individual workplaces can be found within the list area in the order in which they have been assigned to the terminal in MOC.

As regards the basic display of the *AIP* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured within the terminal configuration at the client. The individual basic views are described in the sections that follow.

Footer

The MPDV logo can be found at the bottom left of the screen. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

AIP status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.

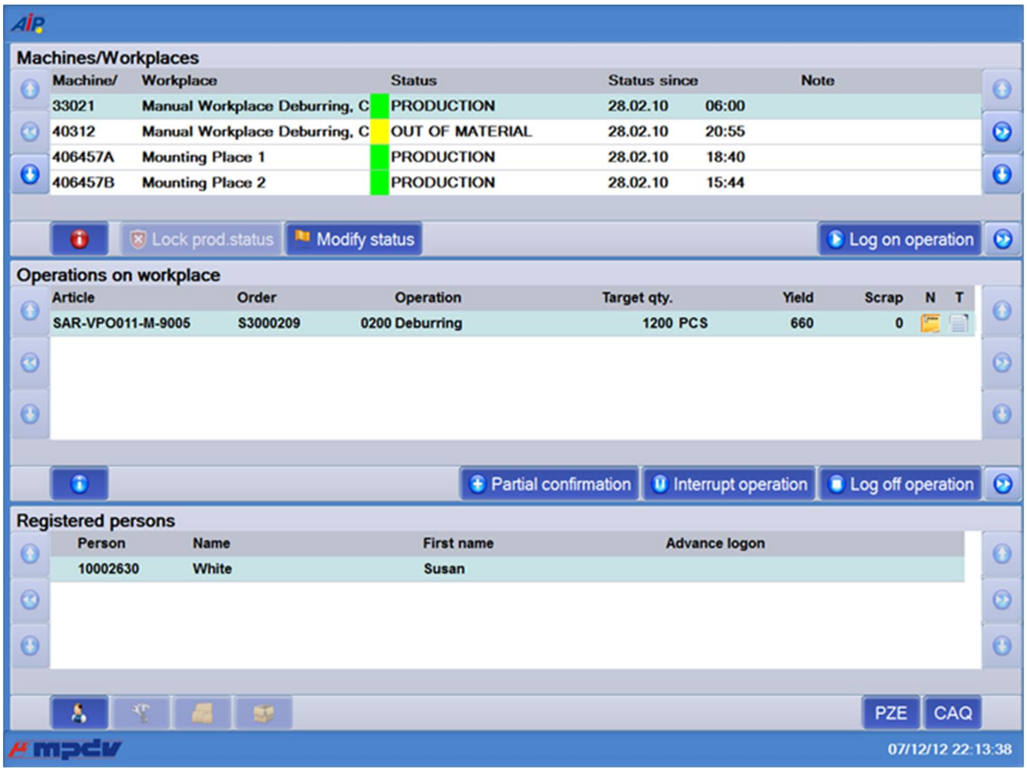
	Commands are being transferred to the server
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

3.2 Basic display “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff



The tabular basic display consists of two or three tables, subject to the configurations made. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional display).

“Machines/workplaces" table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as the designation are displayed.

Status

The status is displayed in color and as status text. Coloring is as follows:

- green: Production
- yellow: Assigned status
- red: Not assigned

Status since

Point in time since the status is available.

For ADE workplaces[1] the point in time refers to the last manual status change, for MDE workplaces it refers to the time when the last status change was identified (for machine connections), to the point in time when the status was changed manually the last time or to the time of the last shift change.

Please note

It is indicated here if the “block production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.

Please note

By way of “customizing” services it is possible to adapt the layout of display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic display of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately, once they have been entered but only once lists have been reloaded.

[1] We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an ADE workplace.

"Operations at workplace" table

The second table shows the operations that are currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number.*

Target quantity

Target quantity that is defined for the operation.

Yield

Yield which has already been produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity which has already been produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note, which should be visible at the terminal, has been recorded for this operation in the graphic planning board at the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a line that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured respectively. Which information is displayed in this list depends, among other things, on the workplace configuration.

The following lists can be displayed:

- Staff logged on to the currently selected workplace (BDE)
- Resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow for switching between these lists.

Please note

The registered staff displayed in the third list correspond to the list of the dialog "F5 registered persons…". Selecting a person in the third list does *not* affect selection of the operation in the list of OPs running at the workplace and, as a result, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic display

A toolbar, which may be configured by customizing, is assigned to each list in the basic display. This makes the purpose of the function clear to the user. The "partial upload/confirmation" function can be found, for example, below the list of registered operations.

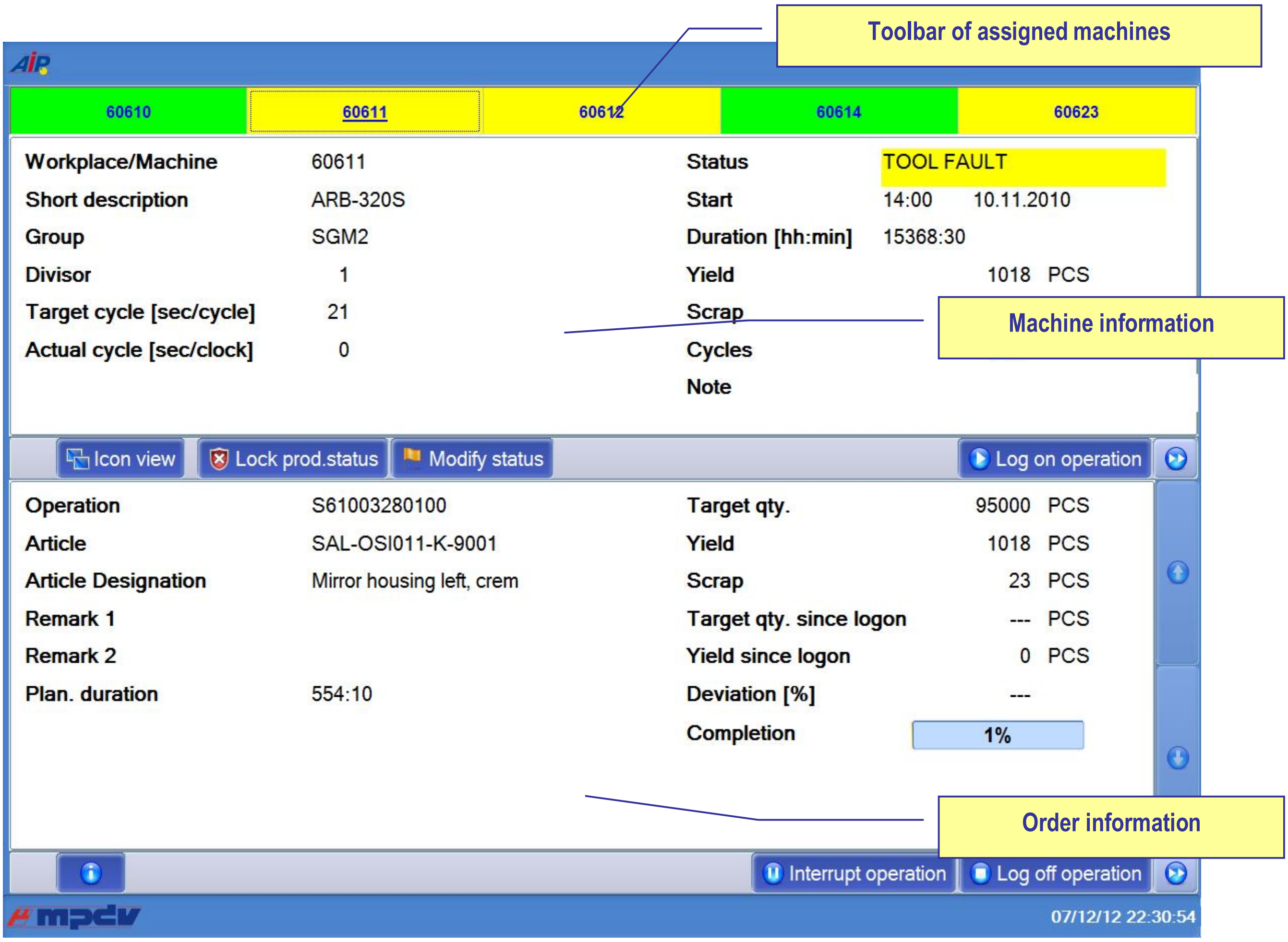
In fact, the toolbar may include several "tabs", which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. change partitioning) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

3.3 "Machine overview" basic display

If the "change view" button is clicked in the basic display, the view changes to the following presentation:



This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation altogether has three areas:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines as well as to shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. Provided that several orders/OPs are logged on, it can be switched between them using arrow keys that are then shown additionally.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$Displayed\ \ partitioning = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + ... \right]$$

TLG_M	Partitioning of the machine (TLG in mnr.lst)
DIV_M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG_{AGi}	Partitioning of the individual order (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview at the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal may get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) at the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time in which the production lock of the machine has not been active). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:
TargetSinceLogon = NetRunningTime[sec] * Partitioning/TargetCycle[sec/stroke]
NetRunningTime: Time since logon, in which the production lock has not been set. This calculation does not take into account any breaks that might be defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

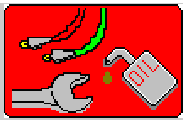
Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (YieldSinceLogon - TargetSinceLogon) / TargetSinceLogon

Completion

The bar represents the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced by a picture showing a yellow or red oilcan, depending on the maintenance activity that is required:



3.4 “Machines as icons” basic display

This view can be configured as the default display within the terminal configuration at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon presentation of machines”.

4 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.

Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

4.1 Basic configuration

Entry	Comment
Section [system]	

Entry	Comment
Section [system]	
`Usr=21`	Distinct terminal number

`Hypath=d:\hydra\`
 	
`Hypath=/usr/hydra/`
	HYDRA server path:
	
	Windows NT:
	-> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed
	Unix:
	->Unix notation

`Hostname=192.9.200.24`	Internet address of the server
`Offlinetimeout=600`	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
`Showcursor=on`	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
`Loadfile=` `ctnet\win\ctaip.txt`	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within “hypath”)
`Watchdog=on`	ON: Watchdog is activated OFF: Watchdog is not activated
`Demo=off`	‘on’: Offline demo mode; always off in the production environment
`parameters=-t`	The –t parameter switches off the virtual keyboard
`TMOUT_C=xxx`	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_S=xxx`	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing

Entry	Comment
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	**Configuration of customized barcode prefixes.**
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
Com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to the connected devices MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
ZAEHLER=|1|2|3|4|5|6|7|8|	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER=|1|2|3|4| IN=|5|6|7|8|
IN=|9|10|11|12|13|14|15|	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
CHARGE=|5|6|	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
MSSZyklusBerechnung=ON MSSZyklusReferenz=0	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
CalculateCycle=ON	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.

Entry	Comment
`WochenEnde_ProdCheck=ON`	This function prevents the weekend automatic from affecting the “production“ status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.
`sFrom999ToNotAttributed=OFF`	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the “no monitoring” option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	

`Button=Editor`
`WindowName=Editor`
`SearchParts=On`
	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button.
	
	Button: button caption
	WindowName: Name of the program (e.g. from the taskbar).
	SearchParts=On: parts of WindowName are sufficient
	SearchParts=Off: WindowName must be entered completely.
	The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.

`ProgFileName=c:\Programme\wincmd\Wincmd32.exe`	The program that is started if the program mentioned above cannot be called to the foreground.
`AutoStart=on`	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
`Modus=PDV` `Modus=PDV,BDE`	Operation as interface IOP←→DOS terminal (standard) Operation as shop floor terminal with PDV
`Terminals=121,122,123`	CTDOS terminals “connected” by LAN. Only required with Mode=PDV
`PDVTerminalDir=c:\hsrv\spool\` `PDVIOPDir=c:\IOPSim\`	Directory for communication with CTDOS terminals Directory for communication with the IOP

`For debug purposes only:`
 	
`InfoFenster=100`
 	
`SlowDown=600`
	
	Number of lines of the “current” window (presentation of last PDV actions)
	
	Slow down, to make events “visible”

`;Supported barcodes`	

Entry	Comment
`FieldWNRBarcodeOnly=Y`	If this entry is set the tool number may only be entered using a scanner.
`FieldNestBarcodeOnly=Y`	If this entry is set the cavity number may only be entered using a scanner.
`FieldNummBarcodeOnly=Y`	If this entry is set the number may only be entered using a scanner.
`FieldKNRBarcodeOnly=Y`	If this entry is set the badge number may only be entered using a scanner.
`BarcodeWNR=`	This field specifies which acronym is entered into the tool number field by the scanner.
`BarcodeNest=`	This field specifies which acronym is entered into the cavity number field by the scanner.
`BarcodeNumm=`	This field specifies which acronym is entered into the number field by the scanner.

5 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version
[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section [section <TNR-USER>]
<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User

The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
`FollowExternStatus=on`	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
`InstallFonts=on`	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
`OnlyInstallFontsAfterDownload=false`	“`InstallFonts=on`”: If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
`InstallTvicport=on`	If “off” the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

AttachedApplication=First	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: **First** → search for connected application first **AfterUserViewer** → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) **Last** → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (**default**). **Off** → Connected application is never searched.
HTTPBrowser=standard	Viewing of documents (via OP info): If documents are configured with a path of the type "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
SupressErrorMessage=70012	Suppress message "material is not planned"
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) **true** → activates keyboard input for the "user" field in the terminal

`Transparency=255`	The signature dialog can also be transparent. **255** → Signature dialog is 0 % transparent (not transparent) **1** → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)
`ShowPosition=TR`	The position of the signature dialog can be adjusted as follows: **TL** Top – Left **TM** Top – Middle **TR** Top – Right **ML** Middle – Left **MM** Middle – Middle (Default) **MR** Middle – Right **BL** Bottom – Left **BM** Bottom – Middle **BR** Bottom – Right Available as of CTAIP (V# 2.0.2.25)
`USE_SERVICE_ACCOUNT=1`	**0** (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO). **Please note:** ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
`SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY`	**REPORTING_USER_READONLY** The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. **REPORTING_USER_CHANGEABLE** The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

SIGNATURE_1_LOGON_TYPE=HYDRA	**“” / Not set / “EMPTY”** There is also an alternative login procedure. **HYDRA** The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. **ACTIVEDIRECTORY** The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. **MIXED_BUT_UNIQUE** Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO“ enabled → Windows only "SSO“ disabled → HYDRA only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures with the terminal in the area of quality data collection.

5.1 Layout configuration

Entry		Comment
Section and/or	**[terminal configuration 0] [terminal configuration 2XXX];**	**(general configuration) (2XXX terminal-specific configuration)**
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE		In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON		As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more

Entry	Comment
	than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;**(general configuration)** [QRD-PRINTER->TICKET 2xxx] ;**(2XXX configuration for a specific terminal)**	
`COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >`	Reloads the machine row for the configured **`<Events>,`** if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
`COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >`	Reloads the order row for the configured **`<Events>,`** if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
`COMPLETE-...-EVENT=< Events >` **`COMPLETE-...-EVENT=#ALL#`** **`COMPLETE-...-EVENT=A_AN|A_P_AN`**	Explanation on the configuration of **`<Events>`** ➔ Using **<#ALL#>** the row (ANR/MNR) that is not available is reloaded for any event. ➔ **<A_AN|A_P_AN>** restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.

6 Local Configuration File ctaiplay.ini

The layout is configured for specific terminals in the file ctaiplay.ini in the terminal directory c:\ctaip.

This file is basically used for the configuration of grids in AIP.

The server directory \hydra\ctnet\win\aip includes complete INI files pertaining to HYDRA standard. Any deviations are developed in specific, customized directories e.g. \hydra\1\custom\aip\tgrp_901.

The relevant, empty file (e.g. ctaiplay.ini) is created here. Modified sections are copied to this file. Then configuration takes place in this file.

After restarting the terminal, files from the main directory \hydra\ctnet\win\aip are merged with files from the customized directory \hydra\1\custom\aip\tgrp_901. Then the merged file is transferred to the local terminal directory C:\aip.

Changes to the configuration file ctaiplay.ini will not take effect until the terminal software has been restarted.

Entry	Comment
Section [OP info]	
Deaktiviert=AG_Bmk,AG_Fort	- indicated info pages are not shown - AG_Info (OP info) cannot be disabled - Entries affected by sorting are not disabled.
Sortierung=AG_TechInfo,*	- Order of info pages in the icon list - if the list ends with " ,* " the non-listed standard pages are added at the end. - Standard pages: AG_Info,AG_ZuInfo,AG_Bmk,AG_TechInfo,AG_Fort,AG_FertPap
Sortierung=AG_Info,AG_AD_Satz,TextFile1,TextFile2,Bild1,* TextFile1=C:\LOGOPAK\druck.txt,druck.txt TextFile2=C:\LOGOPAK\druckerg.txt,druckerg.txt Bild1=C:\aip\spool\screen1.bmp	Configuration of any texts and images that can be displayed in the OP info dialog.
Section [main]	
Nachkommastellen=0	Decimal places for quantities in the order/machine overview
Repaint_time=60	Cycle for updating the view (for machine list and machine info)

Entry	Comment
Buttonkonfigaktiv=0	Reserved
layout=100	Reserved
AuftragZusatzInfoFontSize=12	Font size of texts in the "OP info" dialog
AuftragZusatzInfoFontName=Courier New	Font type of texts in the "OP info" dialog
PopupSize->EmptyQueue=300	Empty popup window size for quick queue
PopupSize->ReloadPze=200	Reload popup window size for PZE configuration
SymbolAdditionalInfo=MBEZK	Display of any field from the machine list in the icon view between machine number and operation number The machine number is replaced by the configured field if lines and aggregates are configured.
SymbolSubstDesignation=MBEZK	The specified field replaces the machine number in the icon view.
Sections for list layouts	
[Personenliste]	List displayed when staff is logged on
[Bedienposition]	Predefined list of "operator positions"
[Maschinenstatusliste]	Predefined list of "machine statuses"
[Ausschussgruende]	Predefined list of "scrap reasons"
[Abweichungsgruende]	Predefined list of "deviation reasons"
[Auftragsliste]	Lower list in the main view
[Vorgabeliste]	Order sequencing list
[Schichtinfo]	List of shift info
[Maschinenliste]	Upper list in the main view
[Eingangsslosliste]	List of input batches, e.g. when "logging on the OP" in batch mode
[Ausgangslosliste]	List of preceding batches when "changing output batches"
Syntax of table formatting	
GRID_FONT=Arial	Font type
GRID_FONTSIZE=9	Font size
GRID_COLOR=clBlack	Font color
GRID_BACKGROUND=clWhite	Background color
GRID_FIXCOLS=0	Number of fixed columns
GRID_ORDER=MSZEIB GRID_ORDER=MSZEIB=-	Sorting Sorting in descending order **Please note:** If several criteria are indicated (separated by |) only the first criterion can be sorted in descending order. All other criteria are sorted in ascending order.
GRID_FILTER=MGRP=0	Table filtering Please note: NOT supported for script dialogs.
GRID_LIST_TYP=MNR GRID_LIST_TYP=ANR	The list type of the section is indicated with this entry, if fields are displayed that need to be loaded additionally. This entry also enables the search when starting. The entry has to be entered above the IDs to be reloaded!!! All IDs that can be reloaded can be found in the file **headers.txt** within the "spool" directory of the terminal. It consists of four lines: 1. Fields that are always included in the machine list 2. Fields that can be reloaded for the machine list 3. Fields that are always included in the order list 4. Fields that can be reloaded for the order list

Entry	Comment
EXAMINE_CONTENTS_CHANGE=MGRP EXAMINE_COLOR_C1=clWhite EXAMINE_COLOR_C2=clSilver	The font color switches from clWhite to clSilver every time the MGRP value changes Up to 8 colors can be defined.
EXAMINE_SCANEXPR1=MGRP=71|72|73 EXAMINE_SCANEXPR2=MGRP=96|97|101 EXAMINE_SCANCOLOR1=ClGreen EXAMINE_SCANCOLOR2=ClRed	The machine groups 71/72/73 are presented in green font color; the groups 96/97/101 are displayed in red font color. Up to 8 colors each can be defined.
EXAMINE_SCANBKEXPR1=BATTRIB=1 EXAMINE_SCANBKEXPR2=BATTRIB=2 EXAMINE_SCANBKCOLOR1=clBlue EXAMINE_SCANBKCOLOR2=clLime	All lines with BATTRIB=1 are shown in blue background color; rows with BATTRIB=2 are displayed in lime. Up to 8 colors each can be defined.
EXAMINE_COLOR=TEXTCOLOR EXAMINE_BKCOLOR=HGRCOLOR	Specification of a column that includes the color value for the row (e.g.: 0-Black; 255-Red, 16777215-White)
EXAMINE_CELLBKLEVEL2=EGR:AUSP,EGR:AUSP,<1*clLime|<=5*clYellow|>15*clRed	Setting of the background color depends on whether the field value reaches different threshold values.
EXAMINE_ROWBKLEVEL1=SGR:REST, SGR:REST,<=0*clRed|<=5*clLime|>15*clYellow	Setting of the background color depends on whether the field value reaches different threshold values. The entire row is colored because of the threshold value.
Syntax of column definitions	
MNR=C8,80,R MGRP=N6,60,R AGR:AUS=N10.2,125,R MSDATB=dd.mm.yy,70,L MSZEIB=hh:mm,70,L SKDATB=dd.mm.yyyy,90,L SKZEIB=hh:mm:ss,80,L MSDAUER=ddd.iii,60,R	Alpha-numeric, 8 characters, 80 pixels, right-aligned Numeric, 6 characters, 60 pixels, right-aligned Decimal, 10 places, 2 decimal places, .. Displayed in the form "23.03.98" (left-aligned) Displayed in the form "08:24" Displayed in the form "23.03.1998" Displayed in the form "08:24:39" Displayed in industrial time unit " 22,982"
AGR:BMK11=hhh:mm:ss,80,R,TESTHEADER	TESTHEADER: new column caption
ALIAS KOPIE=MNR=N8,120,R,TITEL	ALIAS new name is being introduced KOPIE= new identification MNR= ID in data file N8,120,R, Formatting TITEL column caption in table
ALIAS AKA=MNR[1..3]=N8,120,R,ARRAY[1..3]	The first three characters from MNR are output
ALIAS ATTR=MNR(2)=N8,120,R,PARAMETER(2)	The second part separated by „ ; “ is output. Example: „ 12;20;130 “ ➔ „20“
ALIAS U=(R*6.2831)=N10.3,60,L,Scope	Conversion of a value Syntax: see below

Entry	Comment
ALIAS SOLLZ={_INT((_DATETIME(|SKDATE|,|SKZEIE|)-_DATETIME(|SKDATB|,|SKZEIB|))*86400)}=hh:mm:ss,60,R,SOLLZ ;target time of shift	Complex calculations relating to several fields Syntax: see below
GRID_BROWSEROW=0	only the active row is colored yellow
GRID_CELLPAINT=ON	Requirements for coloring rows column by column
GRID_REFRESH=5000	Cycle for updating the display [ms] → lists are not reloaded from the server! Recommended if a constantly changing value is calculated using an ALIAS function.
GRID_POSITION=ON	Display of the grid position  Limited/no support when scrolling using scroll bars and page scrolling
GRID_CELLPAINT=ON EXAMINE_CELLBKCOLOR=WTK:STA,WTK:STA,0-clGreen|1-clBlue|2-clYellow|3-clRed can also be used with index: EXAMINE_CELLBKCOLOR1..8 As of 2.0.2.64: EXAMINE_CELLBKCOLOR1..20	One single column is colored in every cell subject to a value. 1st value: ID of the column to be colored. 2nd value: ID of the reference column 3rd value: Configuration (color for possible values) Please note: - The reference column MUST be shown in the list, if required with length 0 - The values are converted into capital letters when being compared.
As of ctaip V# 2.0.2.17 EXAMINE_CELLBKCOLOR=DMY,COLOR	Take over the color directly from the "color" column. The column <DMY> is shown in the color defined in the column <COLOR>
; Definition virt. column(1) EXAMINE_CELLVALUE1=CV1,REF1,S=DAT|P=ZEI|A=REST|N=INFO ; Definition virt. column(2) EXAMINE_CELLVALUE2=CV2,REF2,10=MGRP|20=MNR|30=COLOR|40=MSTTXT ... ; Layout/Position virt. column(1) CV1=CELLVALUE,150,Z,Data ... ; Layout/Position virt. column(2) CV2=CELLVALUE,150,Z,M/C/T	Filling of a virtual "Case" column with values from different columns subject to the value of a reference column. 1st value: Identification of the virtual "case" column 2nd value: Identification of the reference column 3rd value: Configuration Reference value + ‚=' + display column Please note: the virtual "case" column needs to be configured as follows <Identifier>=<Key word>,<Width>,<Alignment>,<Caption> e.g. CV1=CELLVALUE,150,Z,YST

Entry	Comment
GRID_RANDOMSORT=ON	This options randomly sorts the list. Please note: If this option is active, any configured sorting will be ignored.
GRID_CLIPBOARD=<BUTTON>@<SELECT>@<DATA>	This option copies data from a table/grid into the clipboard.
Special entries	
[Auftragsliste] ALIAS Restlaufzeit=RLZ= C4,44,R,RLZ ALIAS SollErreicht=ZBV1= C1,12,R,S	Activation of the calculation & display of the remaining run time Activation of the calculation & display for target quantity reached (column 'S': '*' if reached)
[Maschinenliste] ALIAS StkProMin=IZYSM= N8,48,R,Stk/min	Activation of the calculation & display of the produced pieces per minute
[layout pze] KundenBitmap=kunde.bmp DienstGangTaste=1,3 StdSchrift=Arial StdDateSize=30 StdStatusSize=26 StdSpdBttnSize=16 InfoSchrift=Courier New InfoSchriftSize=20 SmallStatusFontSize=16 DateTimeLayout=dd.mm.yyy hh:mm:ss	Configuration of the PZE terminal **"Kundenbitmap=<File name>**" file with customer logo When restarting the terminal, this file is copied from the server directory ".\ctnet\win\aip\etc\" into the application directory ".\etc\". "**DienstGangTaste=1,3"** → Default [empty] By entering the function key numbers (1...4), a check specifying if the person is allowed to go on a business trip/ business errand is performed during posting. Configuration of the used font types/font sizes as well as the layout of the date and time display.

6.1 Formulas used in grid layout

Simple conversion of a value

```
Syntax:      ALIAS <Alias>=(<formula>)=formatting
  <formula>: [1/]<KENN>[<Operator><Value2>]
    <KENN>: ID from list (The current value from the list is entered here
            in the formula)
<Operator>: + | - | * | / | ^
  <Value2>: 2nd Operand
```

Extensive formulas:

Formulas that can also relate to several table fields can be recognized by curly braces.

```
 Syntax: ALIAS <Alias>={<Formula>}
 <Formula>: (<Operand1>[<Operator><Operand2>])
 <Operand>: <Value> / <Function> / <Formula> / |ID|
<Operator>: |+|-|*|/|^|
   <Value>: Constant ('0'..'9','e','E','.','-')
    |Kenn|: reads a value from the table
<Function>: _<Fname>(<Operand>[,<Operand>[,...]])
           _DATETIME(<Date>,<Time>)
             <Date>: mm/dd/yyyy
              <Time>: ssss (Seconds of the day (0..86400))
             ==> Real value as TDateTime
           _INT(<Operand>)
             returns a value without decimal places
           _REAL(<Operand>)
             returns a value with decimal places
           _ROUND(<Operand>)
             returns rounded value without decimal places
           _MOD(<Operand1>,<Operand2>)
             ==> Operand1 mod Operand2
           _ABS(<Operand>)
             absolute amount of a figure
           _EXP(<Operand>)
             e raised to the power of X (e: basis of the natural logarithm)
           _LN(<Operand>)
             natural logarithm (Ln(e) = 1)
           _FRAC(<Operand>)
             proportion of decimal places
           _LOG(<Operand1>,<Operand2>)
             LOG(N,X): logarithm to base N of X
           _MAX(<Operand1>,<Operand2>)
             the greater value of two values
           _MIN(<Operand1>,<Operand2>)
             the lesser value of two values
           _SQRT(<Operand>)
             ==> Square root of Operand1
           _PI()
             ==> 3.14151926535...
```

```
Examples:

Calculation of the target time of a shift (ZEISS):
ALIAS SOLLZ={_INT((_DATETIME(|SKDATE|,|SKZEIE|)-
_DATETIME(|SKDATB|,|SKZEIB|))*86400)}=hh:mm:ss,60,R,SOLLZ
```

ALIAS TEST={_INT(|AGR:GUT|/|TLG|)},N3,30,R,Test

ALIAS test1={_LN(2.7182818)}=C8,80,L,Test1

New (V7.2.3.74): Utilization of intermediate variables in ALIAS functions:

ALIAS **U_Brutto**={|AGR:GUT|+|AGR:AUS|}=N8,40,Z,Brutto

ALIAS **U_BPMN**={_REAL(60000/|SZY|)*|TLG|}=N8,35,Z,BpmN

ALIAS TK_TEST={|***U_Brutto**|+|***U_BPMN**|}=N8,35,Z,TK*

Setting of the background color depends on whether the field value reaches different threshold values.

```
Syntax: EXAMINE_CELLBKLEVEL<i>=<Akro>,<Akro_ref>,
                              <Comp1><Val1>*<Col1>|
                              <Comp2><Val2>*<Col2>...
       <i>: Index 1..8
    <Akro>: Identification of the field to be colored
<Akro_ref>: Identification of the reference column
    <Comp>: Limiting characters (<,>,<=,>=)
    <Val1>: Limit value (integer or Real or ID
            of a reference value for comparison purposes)
            alternative: (Akro) – column of limit value
    <Col1>: Color (Delphi name)
```

Threshold values are searched from the left to the right. If a "<“ or a "<=“ – criterion is met, the corresponding color is set and the evaluation/report is finished. If a ">“ or ">=“ criterion is met, it will first be checked whether or not the condition that follows is also met.

The direct comparison with "=“ is not allowed. But the same function can be achieved by processing the comparisons relating to "<“..„<=“ or „>“…“>=“.

An identification put in parentheses may also be indicated instead of the limit value. During the comparison, the current field content including the specified ID is read out from the same row as the limit value.

All three fields (field to be colored, reference field and limit value field, if required) are to be configured as fields to be displayed. The field width can be set to zero if one of these fields should not be visible.

The color value clWhite may be entered to prevent sections from being colored.

The values are compared as they are displayed. The actual values 0.5 and 1 are considered being equal if displayed values are to be rounded to integer values.

Coloring of the field only works if the option "GRID_CELLPAINT=ON“ is set.

The option "GRID_BROWSEROW=0“ should also be set in order for the coloring to be recognized even if the row is selected.

```
Examples:
EXAMINE_CELLBKLEVEL1=MNR,MST,<=1*clLime|<=2*clYellow|>2*clRed
EXAMINE_CELLBKLEVEL2=FS,FS,<90*clLime|>=90*clYellow|>=100*clRed
```

```
EXAMINE_CELLBKLEVEL3=EGR:GUT,EGR:GUT,<(SGR:GUT)*clLime|>=(SGR:GUT)*clYellow
```

6.2 Translations in grid layout

Column contents can be configured to be translated and displayed by entering e.g. the configuration <XYZ=T10,100,L> instead of < XYZ=C10,100,L> in the configured grid columns. A <#> character has to be prefixed for these "resource strings" to provide for better classification. This modification can be used in every INI file (hytnrcfg.ini,..) where grid layouts are configured.

Please note: The data do not include any translated values. In order for them to be displayed in e.g. dynamic dialog fields, an explicit translation has to be performed using the VB script function < vbsTranslateDataValues("<columns>" , "<data row>") >.

Column contents can be configured to be translated and displayed by entering e.g. the configuration <PSPERRE=U1,100,L> instead of <PSPERRE=C1,100,L> in the configured grid columns. The entry for the "resource string" that depends on the field has the following structure:

	„#<Acronym>#<Value>"	
e.g.	**„#PSPERRE#J"**	**"production lock enabled"**
	„#PSPERRE#N"	**„ " (blank character)**

This modification can be used in every INI file (hytnrcfg.ini,..) where grid layouts are configured.

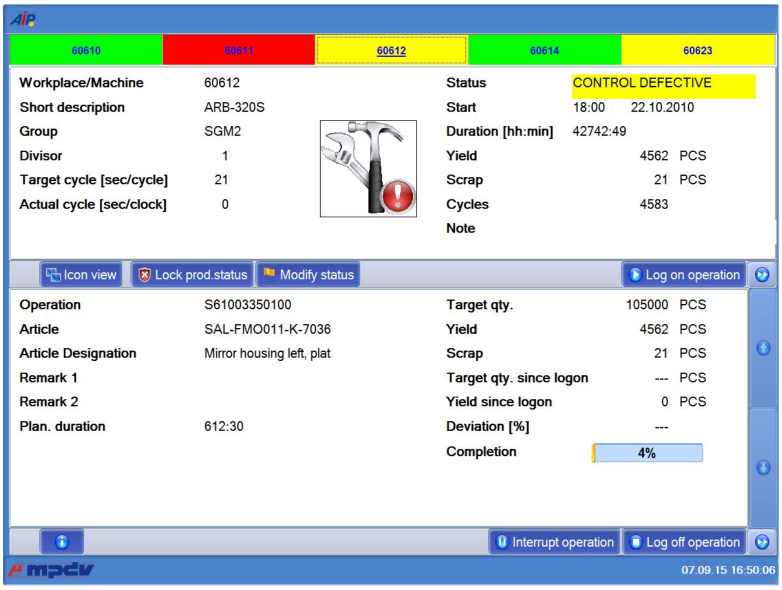
Please note: The data do not include any translated values. In order for them to be displayed in e.g. dynamic dialog fields, an explicit translation has to be performed using the VB script function vbsTranslateDataFields("<columns>" , "<data row>") >.

6.3 Configuration of basic screens

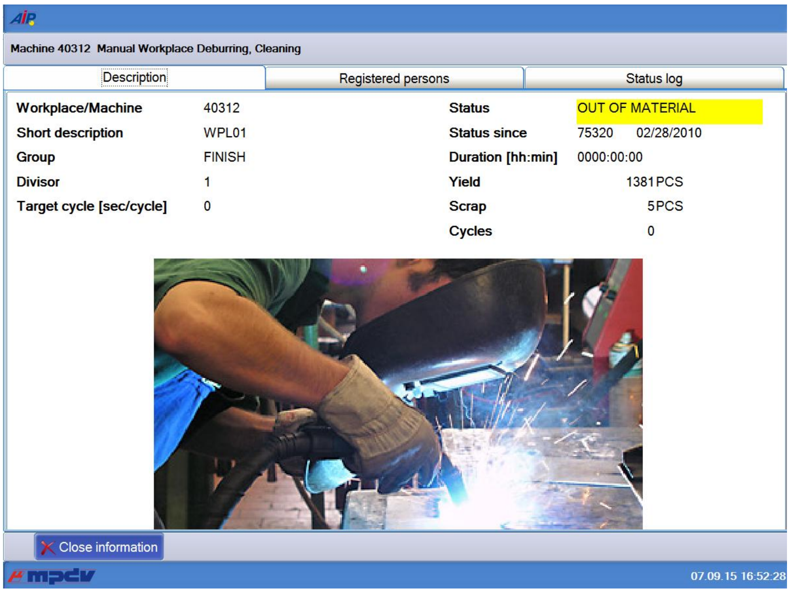
The dialogs/screens are configured using dynamic dialogs. For this reason, the following dialogs are always required:

MMINFO → Section referring to machines in the single machine view

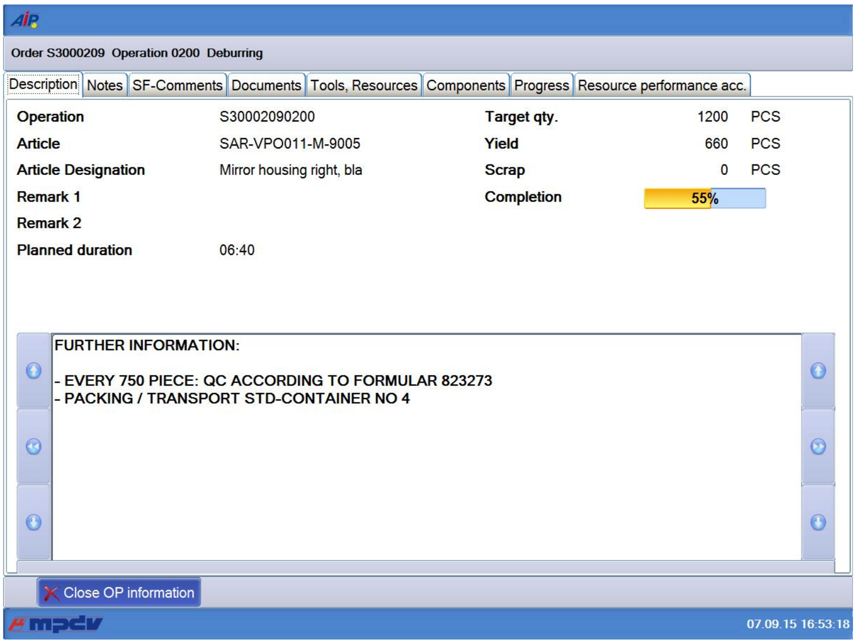
MAINFO → Section referring to orders in the single machine view



MINFO → Description of the machine information



AINFO → Description of the order information



The heights of the individual components of the basic dialogs and, as a result, the positions of the button bar are configured in the *ctaiplay.ini* file using the below-mentioned parameters:

Section [MainView1]	Configuration of the basic screen
MachineGridHeight=415 OrderGridHeight=500 ButtonBarHeight=50	Height configuration of components for the basic screen (machines, order grid, button bar) The configured heights are scaled to the current height. Consequently, the total sum of entered heights does not play a role.
Section [MainView2]	Configuration of the single machine view
MachineGridHeight=50 MachineInfoHeight=415 OrderInfoHeight=355 ButtonBarHeight=50	Single-row grid to select the machine Information on the machine Information on the order Height of both button bars The configured heights are scaled to the current height. Consequently, the total sum of entered heights does not play a role.

6.3.1 Available fields for the dialog configuration of basic screens

A script function that fills the fields according to the customer's requirements is currently not available (14 September 2009).

In general, the fields of the machine list and the order list are available. "MNR." or "ANR." has to be prefixed for identification purposes.

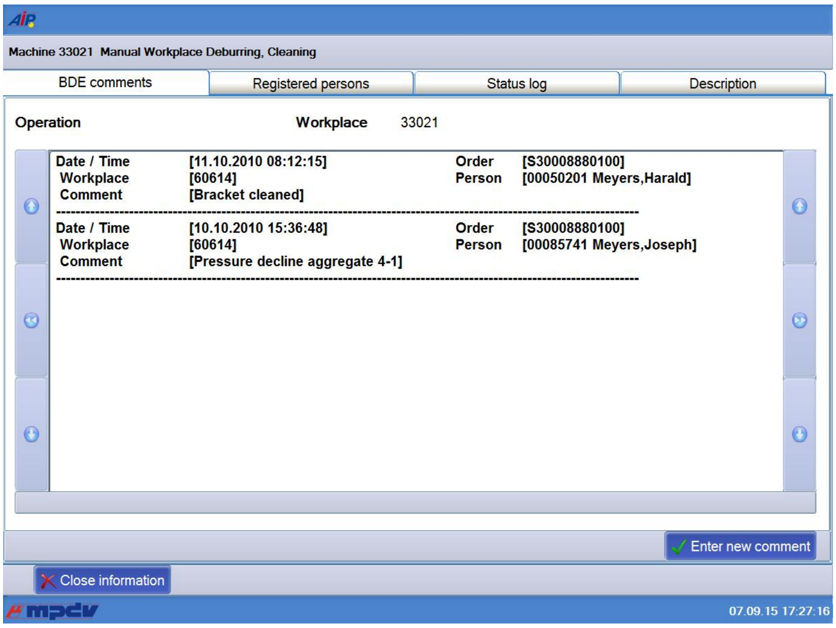
Known quantity fields are formatted to match the configured number of decimal places.

Some fields are calculated. The following fields are additionally available:

ID	Description
ANR.SOLL_SEIT	Target quantity since login The value is determined locally at the terminal. This is only useful for MDE machines. However, the order must be logged on locally after restarting the terminal.
ANR.ABWEICH	Deviation [%] Comparison of "target quantity since logon" and "actual quantity since logon“
MNR.SZY	Target cycle As of AIP 2.0.2.85: Field is transferred including "internal decimal places". The number of characters displayed is determined by the field of the dialog configuration.
MDE.IZY	Actual cycle The machine's current actual cycle - only if MDE processing is active for the machine at this terminal.
MNR.MSZEIB	Start time of the current status
MNR.MSDATB	Start time of the current status
MNR.MSDAUER	Duration of the current status
ANR.BEARBZ	Planned duration
MNR.MSTTXT	Status text
MNR.TLG	Partitioning Calculated based on the orders running at the machine.
ANR.FERTIG	Progress bar
TNRPSPERRE	Placeholder for the production lock (corresponds to the configuration "TNRPSPERRE=U1,150,L,Hinweis“ in ctaiplay.ini) (the value J/N from the list can be found in MNR.TNRPSPERRE)

6.4 Integrate dynamic dialogs in information dialog

Dynamic dialogs can be integrated in information dialogs. The toolbar of the information dialog is then shown below the toolbar of the dialog:



Definition of a dialog for the machine info (ctaiplay.ini):

[M-Info]
;;Dialog<n>=<DLG>,<Tab Caption>
Dialog1=WF_BDE_KOM_CHK,BDE comments
Sortierung=Dialog1,*

Up to 10 dialogs can be configured (Dialog1...Dialog10). The entry "Sortierung=..." causes the tab with the new dialog to be put at the beginning.

Configuration of the toolbar of the info dialog for the dialog (ctaipbut.ini):

[M_INFO.DIALOG1-Page1]
1=MI_CLOSE,L,close machine information

If the "cancel" key is defined in the dialog as described in the above example, it will have no effect here.

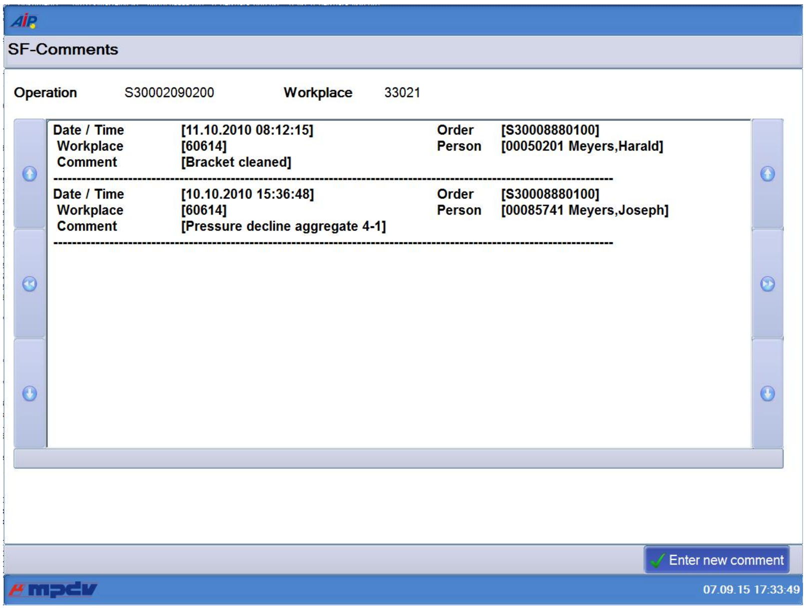
It must be noted that the tabs are displayed over several rows if too many tabs have been inserted. Consequently, the tabs are moved downwards. Relevant areas might reach beyond the display area. To avoid this, the configurable caption texts of the tabs should be as short as possible.

6.4.1.1.1 Restrictions

A dynamic dialog is generated, once it is called. Initially selected data (machine number, order number) are already available.

The dialogs integrated in the information dialogs are already set up when starting the application. Consequently, placeholders included in text fields (e.g. machine <MNR> <MBEZK>) cannot be converted. The function cannot just be started later, as the placeholders have already been overwritten during initialization.

6.5 TextViewer in dynamic dialogs

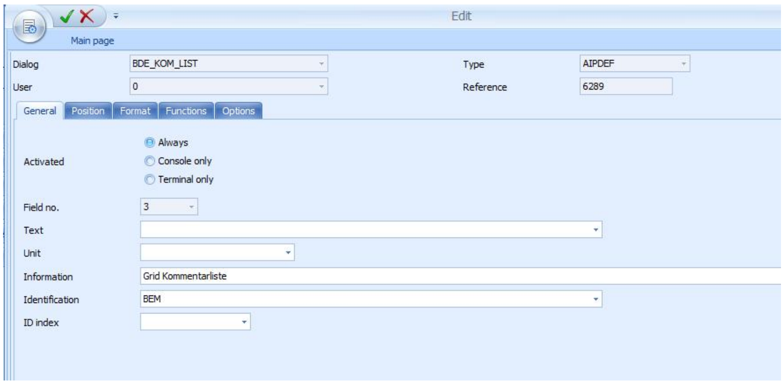


<u>Dialog configuration:</u>

"General" tab:

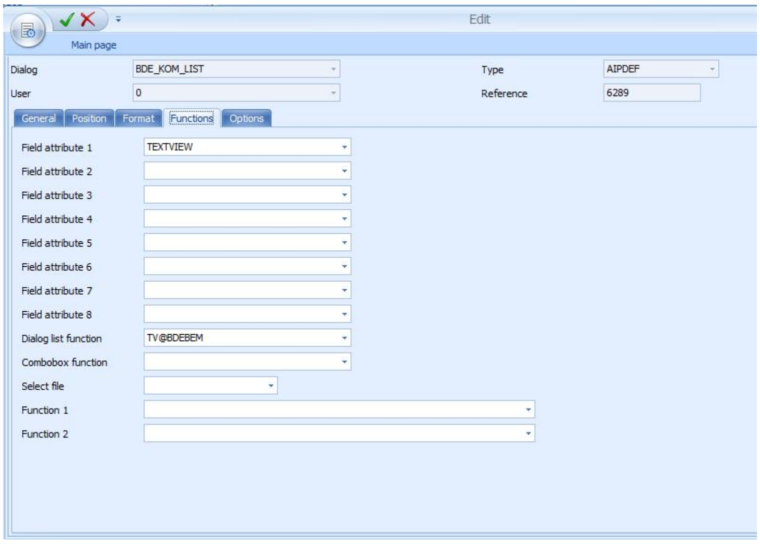
Positioning by field X,Y

Size by unit X,Y



"Format" tab: selection of "grid"

"Functions" tab:



The field attribute "TEXTVIEW“ defines the exact type of the "grid" pre-selection.

The field "dialog list function" includes the section of the ctaiplay.ini file in which further configurations are defined.

Ctaiplay.ini:

```
[TV@BDEBEM]
CMD=DLG=LIST;61|MNR=<MNR>|ANR=<ANR>|
FILE=BdeBem.lst
TEXTFILE=BdeBem.txt
CONVERSION=BDEBEM
FONTNAME=Arial
FONTSIZE=10
```

CMD: Command for loading the list. The placeholders <ANR>, <MNR> are replaced by the current values. The values must be available in the dialog and pre-assigned to default values.

FILE: This file is loaded by the above-mentioned command

TEXTFILE: This file is displayed

CONVERSION: Rule for the conversion from FILE to TEXTFILE (up to now BDEBEM available only). The file is only copied if nothing is entered.

FONTNAME, FONTSIZE: configuration of the font in the TextViewer

7 Local Configuration File ctaipbut.ini

Buttons are configured for specific terminals in the file ctaipbut.ini in the terminal directory c:\ctaip.

The button pages of the main view and the OP info dialog may be configured in the configuration file ctaipbut.ini.

The server directory \hydra\ctnet\win\aip includes complete INI files pertaining to the HYDRA standard. Any deviations from that are developed in specific, customized directories e.g. \hydra\1\custom\aip\tgrp_901.

The relevant, <u>empty</u> file (e.g.: ctaipbut.ini) is created here. All sections e.g. [ANR-ALL-Page1] are copied to this file. Then configuration takes place in this file.

After restarting the terminal, files from the main directory \hydra\ctnet\win\aip are merged with files from the customized directory \hydra\1\custom\aip\tgrp_901. Then the merged file is transferred to the local terminal directory C:\aip.

All sections including the string "page" are imported.

Entry	Comment
Definition of sections [LST-MODUS-PageX.]	**General schematic structure of a button page** **Definition of a section** LST = List identifier of the button page (MNR, ANR, LIST3) MODUS = Mode of the machine LN = MPL – Mode DN = DLL – Mode LR = RF – Mode (reel-based manufacturing) LS = RS – Mode (cutting reels) LC = Handling unit (packing station) or XX = <MPL_MOD>[1] + <TYPE>[] YY = Value from the MNR.LST column <MNRBTN.MODUS> Otherwise if no applicable entry is found for the machine mode, the section ...ALL will be used (if available) X = Button page The definition specifying the mode a machine is running on is implemented in the AIP application program.

Entry	Comment
Sample configuration [MNR-...-Page1] 1=A_AN, L, log OP on 2=BLANK, L 3=$MPL-PAL$PAL_AN,L,log pallet on, 4=%BART:PZE=J%PZE,R,PZE,PZE.BMP Please note: In one section the numbering of entries has to be consecutive 1...n. A gap in numbering indicates the completion of a page!	General structure: x=<Function>,<Alignment>,<ButtonName>,<Icon> e.g. 1=A_AN,L,log OP on,AGAN.PNG - Function A_AN - Alignment L or R (from the first "R“ on always "R“) - ButtonName Log OP on - Icon optional icon name (PNG, resolution 24x24 px) Special functions: $...$ (e.g. $MPL-PAL$) License check fails → Button is deleted %...% (e.g. %BART:PZE=J% .) Check field with value in (T)terminal(K)label ➔ only show if they match BLANK Insert distance between buttons
Configuration of functions using wildcards x=A_AN*,L, log OP on x=A_UN*,R, interrupt OP	The dialog to be opened is located as described below if buttons are configured using wildcards ID A_AN* - calling dialog: A_AN - Determination of the machine type - Supplementing the dialog based on the machine type - Check whether or not the dialog is available if this is the case - calling dialog: A_AN_MPL - evaluation of the posting type (only with A_AN) - Supplementing the dialog based on the *posting type* - Check whether or not the dialog is available if this is the case - calling dialog: A_*P*_AN_MPL - Calling up the located workflow or dialog If the function is <A_UN*> or <A_AB*>, it will be checked whether or not the OP to be logged off is a merged OP. - If this is the case, <A_UN*> is changed into <SA_UN> or <A_AB*> into <SA_AB> If the virtual column <MNRDLG.SUFFIX> includes a value, it will always be used (if available). ButtonFkt MNRDLG.SUFFIX Dialog e.g. A_AN* <XYZ> A(_*P*)_AN_XYZ A_TR* <ABC> A_TR_ABC

Entry	Comment

Further standard buttons:
`P_SPERRE`
`VIEW`
 	
`ICON`
`PDV_ISTW`
`WF_BDE_KOM`
`SA_AN`
`DNC`
`MINIMIZE`
	
	Block production status
	Switching of the basic view:
	List view ←→ presentation of individual machines
	Calling up icon view (only possible if configured in the machine configuration)
	Calling up the actual value view of PDV
	Input of BDE comments
	Log merged operation on
	Calling up the DNC startup screen
	Minimizing of the terminal program (as of V2.0.2.23) → Windows 7 requires the compatibility mode XP

`USER1…USER9`	User-defined buttons to show and start external software The programs are configured in ctaip.ini within the section [ext. software]

Button IDs for the machine info
`[MNR-ALL-Page2]`
`1=M_INFO.INFO,L,show information`
`2=M_INFO.PERS,L,staff`
`3=M_INFO.MSPROT,L,machine status log`
	Consequently, the relevant info dialog including the selected page is opened in the foreground (with focus). Switching to other pages is allowed.
	M_INFO may be used to show the info page in the foreground (with focus):
	M_INFO=M_INFO.INFO

Button IDs for OP info:
`[ANR-ALL-Page3]`
`1=A_INFO.DOKU,L,documents`
`2=A_INFO.HILF,L,production resources and tools`
`3=A_INFO.KOMP,L,components`
`4=A_INFO.BMK,L,RPA`
`5=A_INFO.FORT,L,progress`
`6=A_INFO.NOTE,L,notes`
 	
`A_INFO.TEXT1,L,User text1`
`A_INFO.PICT1,L,User image1`
`A_INFO.SCRINF1,L,User scriptInfo1`
`A_INFO.DIALOG1,L,User dialogs`
	Consequently, the relevant info dialog including the selected page is opened in the foreground (with focus). Switching to other pages is allowed.
	A_INFO may be used to show the information page in the foreground:
	A_INFO= A_INFO.INFO
	
	Direct call of user-defined pages configured in ctaiplay.ini in the section [OP info].
	Example:
	Dialog1=WF_BDE_KOM_LIST,BDE comments
	→ `A_INFO.DIALOG1,L,BDE comments`

Entry	Comment

Configuration of a function with different modes
Just as is the case for the configuration of the ANR/MNR info tabs, a function can be configured with different modes.
`1=RES_WART.MNR,L,machine maintenance`
`2=RES_WART.RES,L,resource maintenance`
`3=RES_WART,L,other maintenance`
	In the configured examples, the dialog < RES_WART > is called with the below-mentioned modes.
	1 < MNR >
	2 < RES >
	3 without mode
	
	The values can be read out as follows in the terminal script.
	VPar(“BTN.FKT“) Function + mode
	VPar(“BTN.FUNC“) Function
	VPar(“BTN.MODE“) Mode

Available button sections and buttons for pages of the OP info dialog:
 	
`[A_INFO-Page1]`
 	
`[A_INFO.DOKU-Page1]`
`3=AI_VIEW,R,open document`
`4=AI_VIEW_CLOSE,R,close document`
 	
`[A_INFO.HILF-Page1]`
 	
`[A_INFO.KOMP-Page1]`
 	
`[A_INFO.BMK-Page1]`
 	
`[A_INFO.FORT-Page1]`
 	
`[A_INFO.NOTIZ]`
 	
`[A_INFO.DEFAULT-Page1]`
 	
Recommended for all pages:
`1=AI_CLOSE,L,close OP information`
	Overview
	
	Document view
	
	Production resources and tools
	
	Components
	
	Resource performance accounts
	
	Progress bar
	
	Notes (as of ADE 7.3)
	
	Configuration of a default page (is used if no section is defined for the tab).
	
	The IDs may also be used for the keys within the dynamic dialog (field "function").

Entry	Comment
Available button sections and buttons for pages of the machine info dialog:	
`[M_INFO-Page1]`	Overview

`[M_INFO.PERS-Page1]`
`2=P_AN,R,log person on`
`3=P_AB,R,log person off`
`4=P_AAB,R,log everyone off`
	Staff logged on

`[M_INFO.MSPROT-Page1]`	Machine status log
`[M_INFO.DEFAULT-Page1]`	Configuration of a default page (is used if no section is defined for the tab).
For all pages: `1=MI_CLOSE,L,close machine information`	

Entry	Comment
Definition of sections **[ButtonPanel]**	**General section for the configuration of global settings for all used button panels** **→ 2 in main view (MNR , ANR)** **→ (W)ork(F)low**
functionkey_visible=on	Shows function keys (e.g. "F3") in button panels in order for the selection to be performed using function keys (default = off).
radiobuttonkey_visible=on	Presentation of function keys in radio group boxes of a workflow (default = off).
functionkey_pze_visible=on	Display of function keys in PZE module (default = off).
Definition of sections **[LIST3-ALL-Page1]** **as of CTAIP V# 2.0.2.33** **..=~<VISLIST-ID>~,L,,<BITMAP>** The characters "~“ (or previously "§“, should no longer be used) have been designed to identify third list buttons. Correct processing/updating (disabled/enabled) is only given in the third grid list of the main screen.	**General section for the configuration of functions of the configurable third list of the main screen.** INFO: The different types of the "3rd list" are configured in the machine label. The layout of a "3rd list" is defined in the "hytnrcfg.ini". ➔ All used lists have to be configured with their identifier „“ as follows. ➔ When changing machines, the "3rd list" may be hidden/shown and buttons for the "3rd lists" that are not configured may be disabled, if necessary.
1=~**M**~,L,,**PALETTE20x20.BMP**	Entry for "material list" ➔ "[VISLIST3(**M**)]“ from "hytnrcfg.ini“
2=~**P**~,L,,**PERSON20x20.BMP**	Entry for "list of persons“ ➔ "[VISLIST3(**P**)]“ from "hytnrcfg.ini“
3=~**R**~,L,,**RESS20x20.BMP**	Entry for "MNR_AMAT.LST“ ➔ "[VISLIST3(**R**)]“ from "hytnrcfg.ini“ with the configured Bitmap „“
4=~**A**~,L,,**NUM.BMP**	Entry for "material list" ➔ "[VISLIST3(**A**)]“ from "hytnrcfg.ini“
5=~**G**~,L,,**PERSON20x20.BMP**	Entry for "list of persons GWP“ ➔ "[VISLIST3(**G**)]“ from "hytnrcfg.ini“

8 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
--------	--------------------	**---- General ---**
00.	00.ABC123	Data not defined, ➔ 00. will be deleted and data “ABC123” will be passed to standard processing
01.	01.OK 01.ESC	Action barcode ➔ Dialog cancelled or ended with OK button or Esc button.
--------	--------------------	**---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---**
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Suborder number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16.200	Machine → Acronym <MNR> ➔ Passed to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> ➔ Passed to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> ➔ Passed to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > ➔ Passed to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> ➔ Passed to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
--------	--------------------	**----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---**
40.	40.100 40.MONTAGE	Destination → Acronym <ZLO> ➔ Passed to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> ➔ Passed to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→➔
43.		Throughput batch number → Acronym <DLL>→➔
44.		Alternative batch number → Acronym <CNR:ALT1>→➔
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
--------	--------------------	**---- Mainly for the HYDRA-PZE module ---**

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> ➔ Passed to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> ➔ Passed to dialog with FGR=1 or FGR=1001
--------	--------------------	---- Customer-specific barcodes ---
90.		
…		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]
BarcodePrefixChar=$

If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font “Codedreineun” and prefix “.“

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

8.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR **BarKenn91=EGR:GUT**	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:
10:ANR
11:AUNR
12:AFOLG
13:AGNR
14:UAGNR
15:RMNR
16:MNR

17:MST
18:AUSGRD
19:AGGGRD
20:BPOS
21:LPKZ
22:SPLNR
40:ZLO
41:TPE
42:CNR
43:DLL
44:CNR:ALT1
45:CNR:ALT2
46:CNR:ALT3
47:CNR:ALT4
48:CNR:ALT5
49:CNR:ALT6
50:KNR
51:PNR
52:KST
53:FGR

9 Extended Application Configuration

9.1 Overview of INI configuration files

INI file	Configurations
ctaip.ini	host name, terminal number, virtual keyboard (on/off), inputs/outputs
ctaip.mld	Translation (different languages)
ctaipbut.ini	Configuration of buttons: order, positioning, icons, if necessary license
ctaiplay.ini	Configuration of grid layout; basic screen: height of tables and buttons; layout of BDE comments; OP info, machine info
dialog.ini	Configuration of the font type/size in dialogs and tab sizes in workflow dialogs
keyboard.ini	Configuration of virtual keyboard: positioning, customized keys

9.2 General

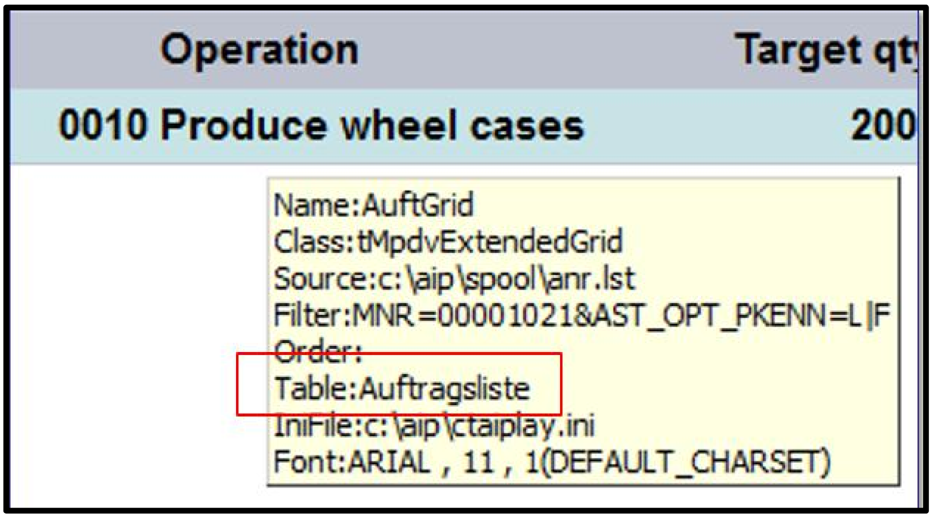
9.2.1 Identification of lists / elements at the terminal

The following keyboard shortcut can be used to show information about available lists or elements at the terminal:

CTRL + ALT + F6
or in
AIP DEBUG menu: Further debug functions → Activate hints (scroll down)

A tooltip is shown when moving the mouse over a table or element.

The value "table" identifies the list:



9.3 Modifications to ctaipbut.ini

9.3.1 General

The server directory \hydra\ctnet\win\aip includes complete INI files pertaining to the HYDRA standard.

Any deviations are developed in specific, customized directories. e.g. for customized terminal groups \hydra\<instance>\custom\aip\tgrp_xxx

(xxx = number of terminal group)

The relevant, <u>empty</u> file (ctaipbut.ini) is filed in these directories. All sections e.g. [ANR-ALL-Page1], which have been customized, are copied to this file. Then this file can be configured.

After restarting the terminal, files from the main directory \hydra\ctnet\win\aip are merged with files from the customized directory (e.g. \hydra\<instance>\custom\aip\tgrp_xxx). Then the merged file is transferred to the local terminal directory C:\aip.

The directory \hydra\ctnet\win\aip must not be changed, otherwise AIP might no longer work properly. In addition, default files are stored there and any changes that might have been made will be lost after updating (e.g. service pack)!

9.3.2 Modifications to the toolbar

A ctaipbut.ini file including the modified section is created in the customized terminal directory (e.g. if the toolbar is to be changed for terminal groups \hydra\<instance>\custom\aip\tgrp_xxx\).

Example:

On the AIP basic screen the order of buttons should be changed in the "machines" section (position "change status" button in front of "lock production status").

Original configuration

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
```

New configuration

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=M_MST,L,Status ändern,Status Flag Yellow.png
4=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
```

9.3.3 Modifications to button labeling

A ctaipbut.ini file including the modified section is created in the customized terminal directory (e.g. if button labeling is to be changed for terminal groups \hydra\<instance>\custom\aip\tgrp_xxx\).

If a customized ctaipbut.ini file already exists, e.g. due to changes to the order of buttons, the required sections can also be inserted.

Example:

Labeling of buttons in the "operation" section should be changed as follows:

- "Partial confirmation" --> "Part. conf."
- "Interrupt operation" --> "Interrupt OP"
- "Log off operation" --> "Log off OP"

Original configuration

```
[ANR-ALL-Page1]
1=A_INFO,L,,InfoBlue.png
2=A_TR,R,Teilrückmeldung,SyBluAdd.png
3=A_UN*,R,Arbeitsgang unterbrechen,SyBluPau.png
4=A_AB*,R,Arbeitsgang abmelden,SyBluStp.png
```

New configuration

[ANR-ALL-Page1]
1=A_INFO,L,,InfoBlue.png
2=A_UN*,R,AG unterbrechen,SyBluPau.png
3=A_TR,R,Teilrück.,SyBluAdd.png
4=A_AB*,R,AG abmelden,SyBluStp.png

9.3.4 Modifications to icons

General

Copy the file pic.zip from the folder \hydra\ctnet\win\aip to the customized directory, e.g. \hydra\<instance>\custom\aip\tgrp_xxx\

Then rename the file *pic.zip* as *pict_cust.zip* in the customized directory.

Please note:
The file pic.zip includes all icons that can be used for the terminal.

The file name for the button icon is to be changed in the customized section of ctaipbut.ini.

Example:

Changing the icon for the button "log on operation" from "SyBluPly.png" to "Tools.png"

Original configuration

[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png

New configuration

[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,Tools.png

9.4 Modifications to ctaiplay.ini

9.4.1 General

The server directory \hydra\ctnet\win\aip includes complete INI files pertaining to the HYDRA standard.

Any deviations are developed in specific, customized directories.
e.g. for customized terminal groups \hydra\<instance>\custom\aip\tgrp_xxx

(xxx = number of terminal group)

The relevant, empty file (ctaiplay.ini) is filed in these directories. All sections e.g. [ANR-ALL-Page1], which have been customized, are copied to this file. Then this file can be configured.

After restarting the terminal, files from the main directory \hydra\ctnet\win\aip are merged with files from the customized directory (e.g. \hydra\<instance>\custom\aip\tgrp_xxx). Then the merged file is transferred to the local terminal directory C:\aip.

The directory \hydra\ctnet\win\aip must not be changed, otherwise AIP might no longer work properly. In addition, default files are stored there and any changes that might have been made will be lost after updating (e.g. service pack)!

9.4.2 Inserting user fields in table

A ctaiplay.ini file including the modified section is created in the customized terminal directory (e.g. if button labeling is to be changed for terminal groups \hydra\<instance>\custom\aip\tgrp_xxx\).

ctaiplay.ini is configured in two steps:

- The field that should be activated has to be defined in the relevant section, e.g. [User fields ANR]
- The field that should be displayed in the grid has to be configured in the relevant section (e.g. to display an additional field in the order list),
  e.g. [order list]

Example:

User field 65 should be named "article name 2" and added to the order list.

Step 1 ➔ define the field in section [User fields ANR]

```
[ User fields ANR ]
GRID_LIST_TYP=ANR
; additional fields in order list
ANR_FU_65= ; Benutzerfeld 65 Arbeitsgang, Artikelbezeichnung 2
```

Step 2 ➔ insert the field for the grid [order list]

```
[Order list]
GRID_FONT=Arial                ; font type
GRID_FONTSIZE=9                         ; font size 9;
GRID_COLOR=clBlack         ; font color
GRID_BACKGROUND=clWhite           ; background color
GRID_LIST_TYP=ANR
EXAMINE_BITMAP1=B1,OPT_INFOAN,T=Attach Notes.png
EXAMINE_BITMAP2=B2,OPT_INFOAI,T=Text Document.png
ATK=C25,100,L,Artikel
ANR_FU_65=C30,150,L,Artikelbezeichnung 2
```

Explanation: SYNTAX of order list

ANR_FU_65 = user field 65
C30 = alphanumeric field with 30 characters
150 = number of pixels for the column
L = left-aligned
Artikelbezeichnung 2 = column caption

9.4.3 Change order of columns in AIP

A ctaiplay.ini file including the modified section is created in the customized terminal directory (e.g. if button labeling is to be changed for terminal groups \hydra\<instance>\custom\aip\tgrp_xxx\).

If a customized ctaiplay.ini file already exists, e.g. due to changes to the order of buttons, the required sections can be inserted there.

Example:

The "order" column should be displayed in front of the "article" column.

Original configuration

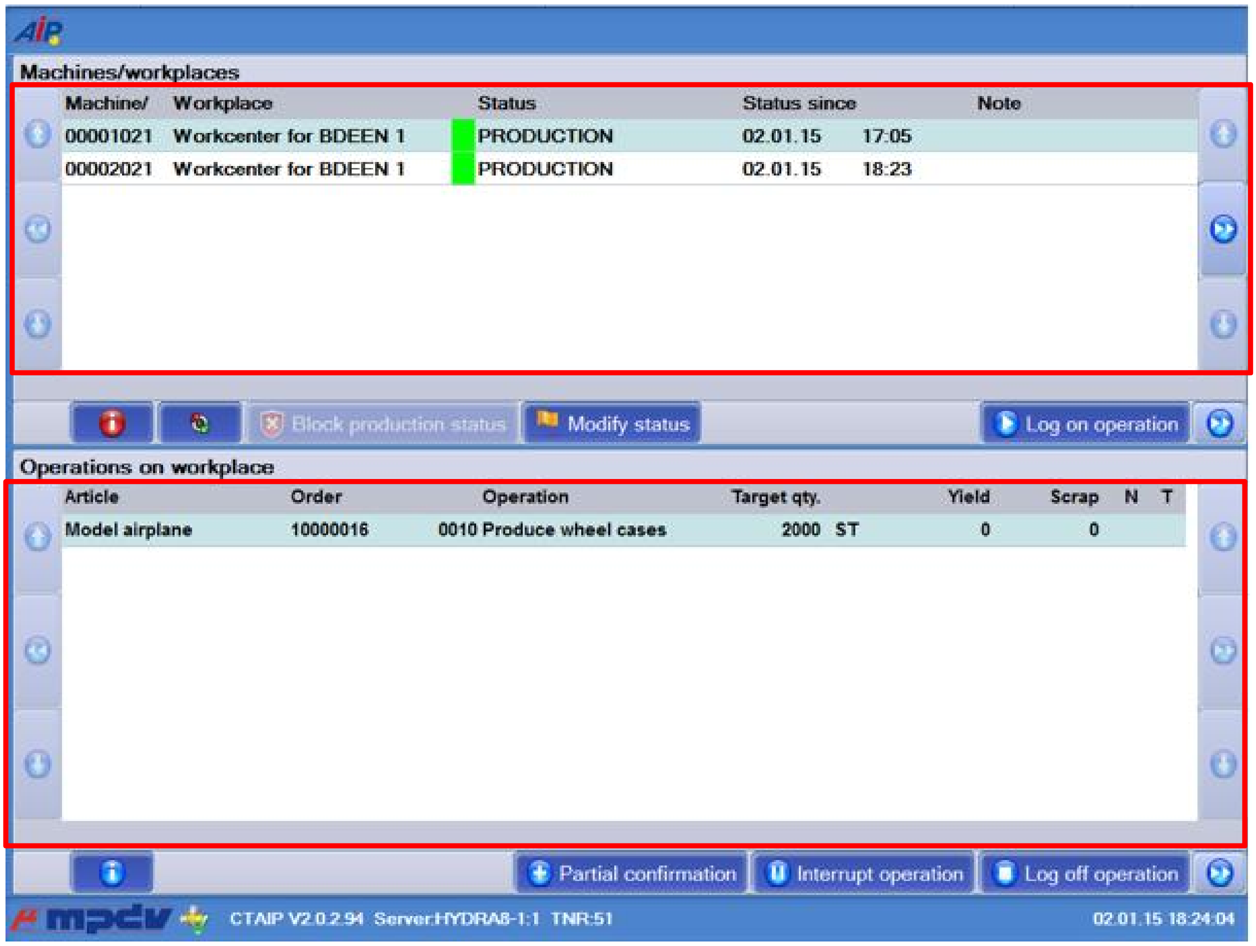
```
[Order list]
…
ATK=C25,100,L,Artikel
AUNR=C10,85,L,Auftrag
ANR_FU_65=C30,150,L,Artikelbezeichnung 2
AGNR=C4,39,R," "
```

New configuration

[Order list]
…
AUNR=C10,85,L,Auftrag
ATK=C25,100,L,Artikel
ANR_FU_65=C30,150,L,Artikelbezeichnung2
AGNR=C4,39,R," „

9.4.4 Changing the height of AIP lists

Changing the height of lists (operation and machine list) in the basic screen [MainView1] of AIP.



The configured heights are scaled to the given height. Consequently, the total sum of entered heights does not play a role.

The heights of components of the basic screen (machines, order grid, 3rd list, toolbar) can be configured in section [MainView1] of the customized ctaiplay.ini file.

```
[MainView1]
; Values are scaled to match the current resolution
; in order to use the full screen. Percentage values can also be entered
OrderGridHeight=440
MachineGridHeight=380
;List3GridHeight=200
ButtonBarHeight=50
```

Explanation: SYNTAX of order list

OrderGridHeight = height of the order list (indicated in pixels)
MachineGridHeight = height of the machine list (indicated in pixels)
List3GridHeight= height of the third list (tools, staff, batches,…) (indicated in pixels)
ButtonBarHeight= height of the toolbar (indicated in pixels)

9.5 Modifications to ctaip.ini

9.5.1 General

The file ctaip.ini is not merged with a file stored in the server. The file must be edited locally in terminal.

9.5.2 Displaying the actual machine cycle in AIP

1. If necessary, create the relevant section [MSS-INIT]
2. Create INI option to call up the function calculating the actual cycle "CalculateCycle=ON"

Example of the section in the terminal file ctaip.ini:

```
[MSS-INIT]
CalculateCycle=ON
```

9.5.3 Start Third-Party Application from AIP

Starting a third-party application is configured as follows:

- Configure a relevant button in the ctaipbut.ini file
- Configure the function

AIP allows for the integration of buttons starting third-party applications in all toolbars. These buttons

- start third-party applications provided they are not running
- call third-party applications to the foreground when they are currently running

Configuration of buttons

The first button that may be used to call up third-party software is configured as "USER1" in ctaipbut.ini.

Further buttons starting third-party software can be configured as "USER2" until "USER9" in the ctaipbut.ini file.

Example:

Configuration of a new button. The button is to be displayed for a specific terminal group in the "machines" section of the AIP basic screen. This can be configured in the ctaipbut.ini file applicable for the specific terminal group in the server.

\hydra\<instance>\custom\aip\tgrp_xxx\ctaipbut.ini)

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; Switching the basic screen between list view and presentation of individual machines:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
6=USER1,R,Notepad
```

Configuration of the function

The function is configured in section [ext. software] of the local terminal configuration file "ctaip.ini":

Example:

```
[ext. software]
Button=Notepad
WindowName=Notepad
ProgFileName=C:\Program Files (x86)\Notepad++\notepad++.exe
SearchParts=On
```

Please note:
"**SearchParts**=“ ➔ If this entry is set, it is sufficient to enter the program name only partly in WindowName.

9.5.4 Remember staff badge number

When logging on an order, the person suggested by the terminal is only changed if the respective person logs on along with the order. (A_P_AN instead of A_AN)

The stored person is removed from memory as soon as he/she has explicitly logged off from the machine (the same is true for "log off everyone").

This default assignment can be avoided by configuring "Default=0" and setting the field attribute SETVALUE in the dialog configuration.

This default assignment occurs with all order postings, when statuses change and batches are posted C_UMB, C_GEN, CA_WL

The stored staff is deleted for all machines when shifts change and/or at the beginning of a shift (Please note: If shifts change for one machine connected to the terminal, staff of all other machines pertaining to this terminal will be deleted as well)!

This can be configured via the entry "HoldPersonInfo=on" in section [SYSTEM] of the ctaip.ini file.

Example:

```
[System]
…..
HoldPersonInfo=on
```

9.6 Virtual keyboard

9.6.1 Change / hide virtual keyboard

The QWERTZ layout can be configured for the virtual keyboard by entering "KEYMODE=QWERTZ" in section [User] of the keyboard.ini file.

Example:

```
[User]
ViewKeysAlways=on
;AlignSize=on
; Show shift key
;ButtonShift=ON
; Disable automatic switching to numeric keyboard
;ContextSensitive=OFF
; Duration of hiding keyboard in sec
;HideTime=10
KEYMODE=QWERTZ
```

The virtual keyboard can also be hidden/disabled if it is not required as the terminal is connected to a keyboard. This can be configured in section [SYSTEM] of the local ctaip.ini file.

Example:

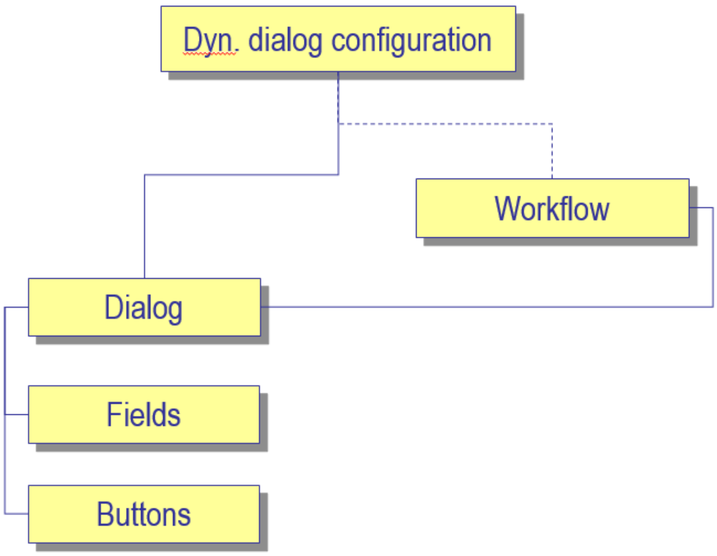
```
[SYSTEM]
Parameters=-t
```

Syntax:

+t/-t --> enables/disables the virtual keyboard; irrespective of the terminal type)

9.7 Dynamic dialogs

9.7.1 Overview



9.7.2 AIP dialog types

AIP provides the following dialog types:

- AIPDEF – Default dialogs (customizing)
- AIPTGRP – Dialogs for specific terminal groups (configuration)
- AIPTNR – Dialogs for specific terminals (customizing)

Only dialogs specific to terminal groups may be created / changed.
Existing terminal-specific dialogs may be changed.
But default dialogs **cannot** be changed.

The terminal has to be rebooted after having made changes to a dialog.

9.7.3 Dialogs for specific terminal groups

It is possible to perform configurations for specific terminal groups.

Before starting the configuration, a backup copy of the concerned dialogs should be made in a backup group (e.g. AIPTGRP 999). (In case old backups exist, delete them beforehand).

The newly added dialogs must be activated and the terminal must be rebooted.

Example:

Special posting dialogs should be used for a terminal group xxx. The workflows and dynamic dialogs are assigned to this terminal group.

Procedure

1. Assign terminal to terminal group
2. Copy workflows to terminal group xxx
3. Copy dynamic dialogs to terminal group xxx
4. Activate dialogs

Assign terminal (MOC)

Menu: System administration --> Terminals --> Terminal groups

Copy workflows (MOC)

Menu: System administration --> Terminals --> Workflow

Copy complete workflow configuration from AIPDEF 0 to AIPTGRP xxx.

Copy dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

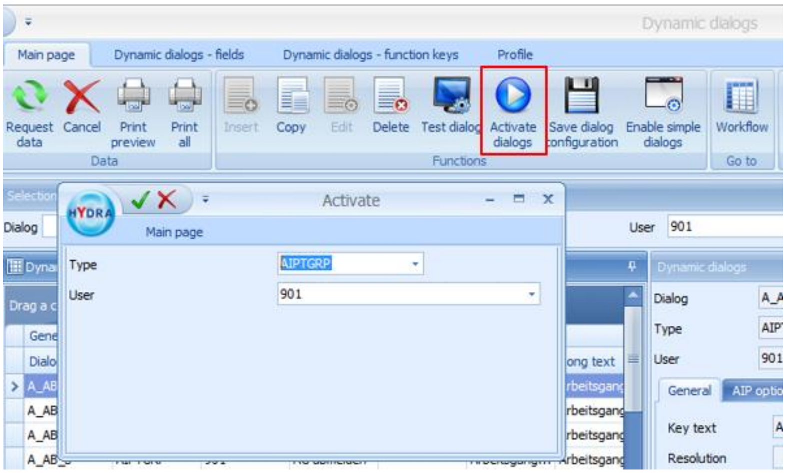
Copy the complete dialog configuration from AIPDEF 0 to AIPTGRP xxx

Activate dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Button "Activate dialogs"

Dialog input: Type =AIPGRP ; User=xxx



Please note:

All dialogs of a terminal group may be deleted at once by selecting all rows of the terminal group (AIPTGRP).

9.7.4 Hide fields (for specific terminal groups)

Identify the dialogs used in AIP

Using the shortcut Ctrl + ALT + F6 a tooltip indicating the dialog name is shown.

General procedure:

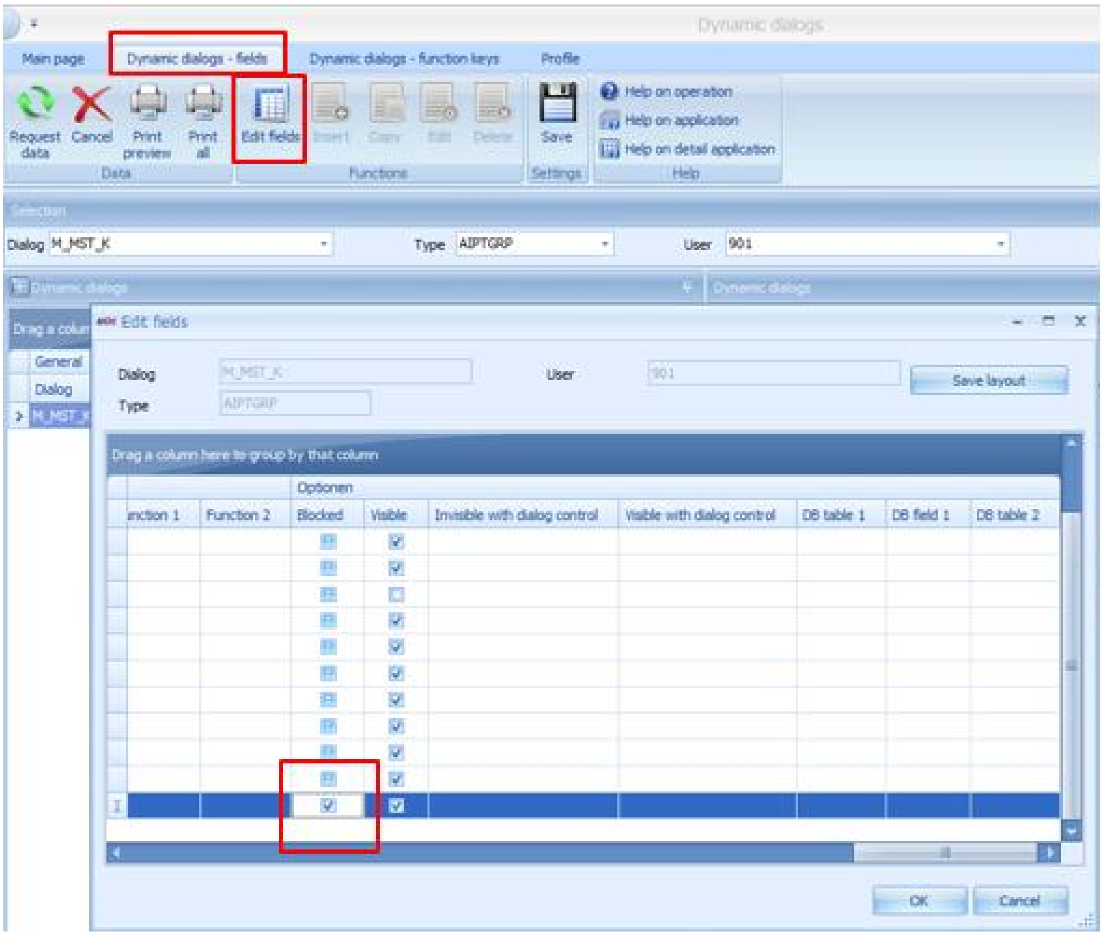
- Identify the dialog in which a field should be hidden
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialog (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

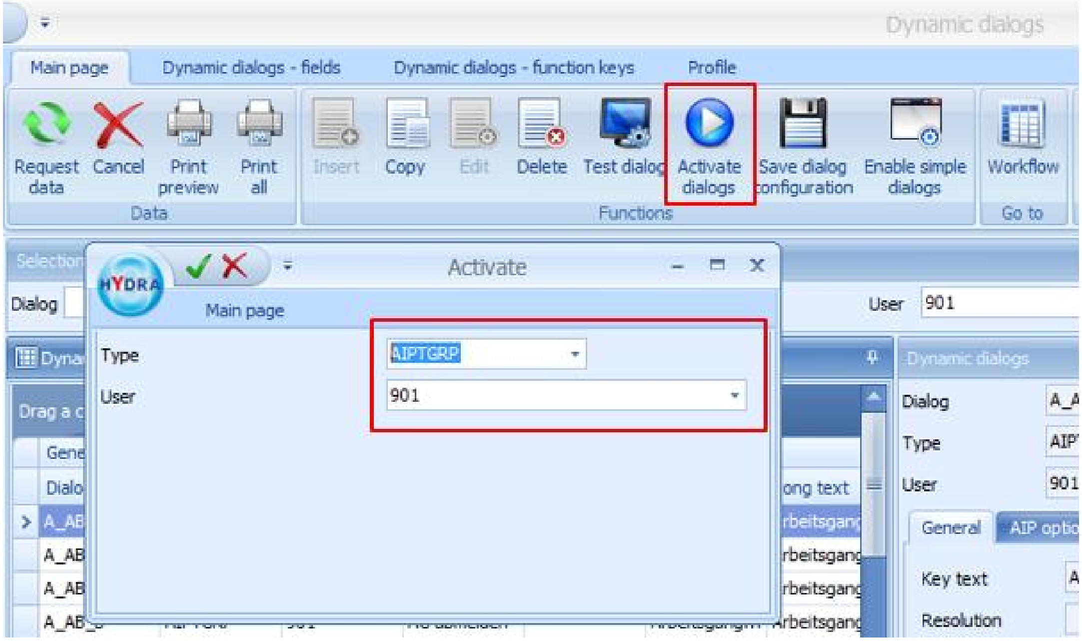
Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Choose the required field and check the option "blocked"



Activate dialog (MOC)

Activate dialog for the specific terminal group



9.7.5 Default assignment in dialog fields (for specific terminal groups)

General procedure:

- Identify the dialog in which a field should be assigned with default values
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialog (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Set field "field attribute 2" to "SETVALUE"

Add field "default"

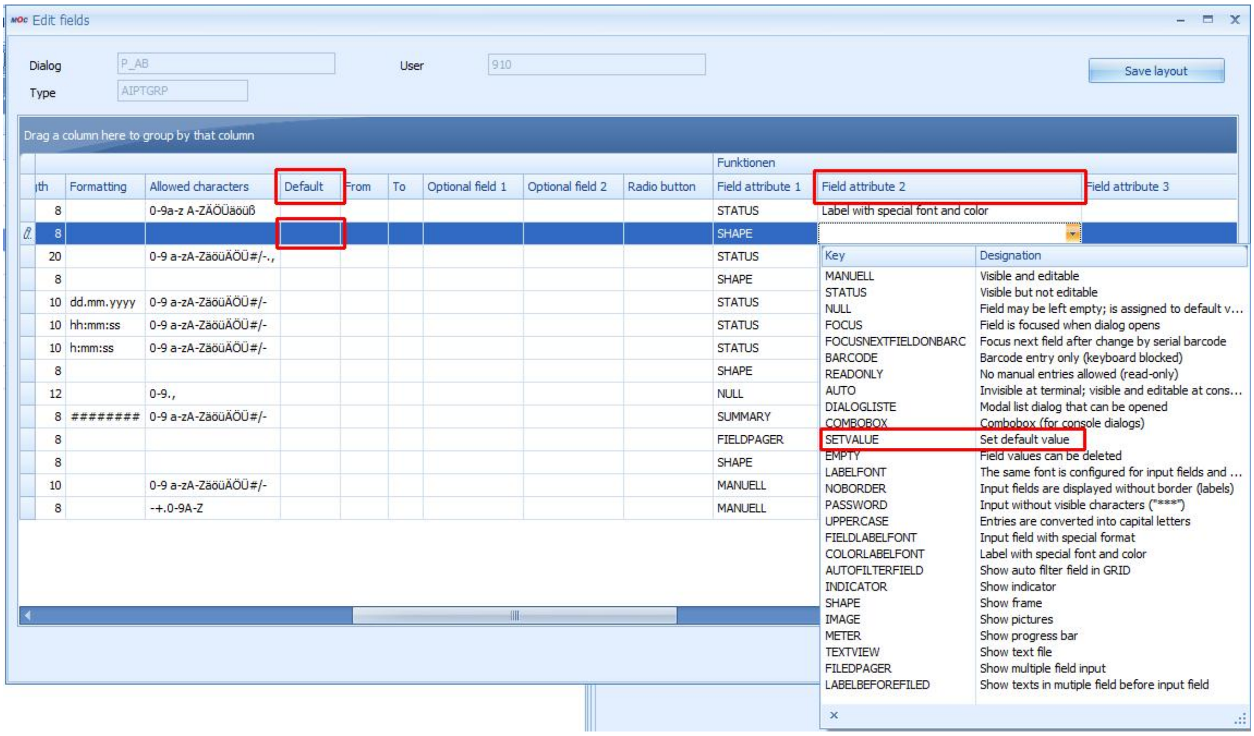
Allowed characters for dynamic dialog fields

The minus character "-" must always be placed at the end to prevent it from being mistaken for the character used for the definition of "from" - "to" ranges.

Example:

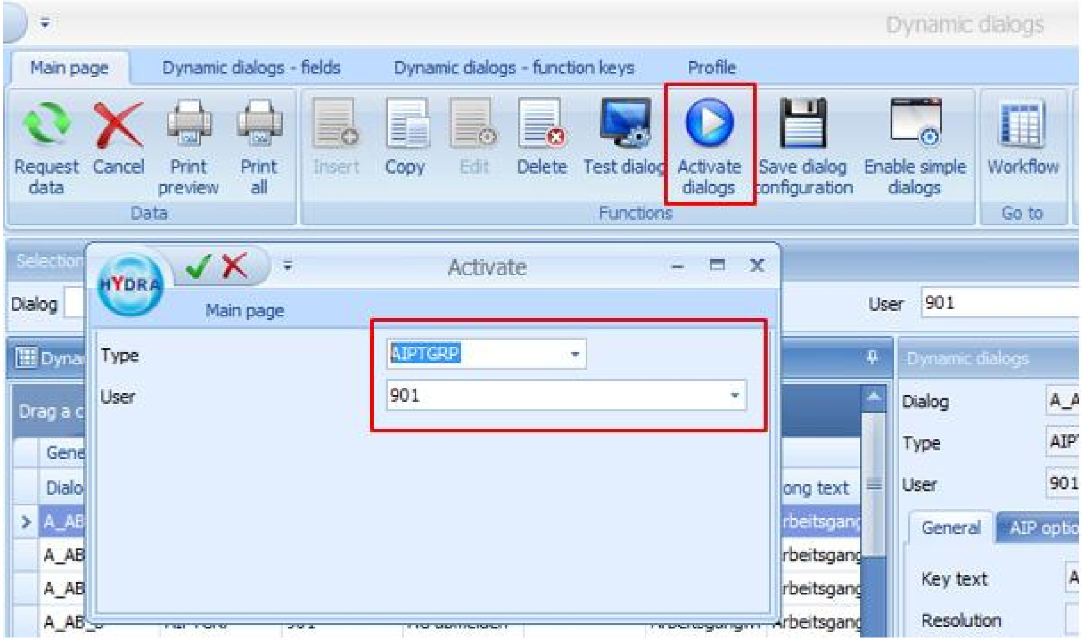
a-z A-Z/-,. is interpreted as range from a to z and A to Z but in this case also as from "/" to ","

a-z A-Z/,.- is interpreted as range from a to z and A to Z and as the allowed characters / , . and -



Activate dialog (MOC)

Activate dialogs for specific terminal groups



9.7.6 Change field name (for a specific terminal group)

General procedure:

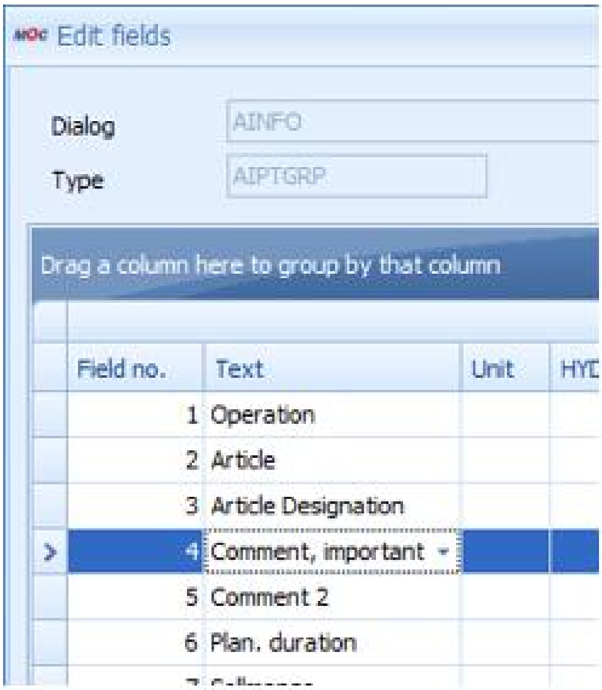
- Identify the dialog in which a field name should be changed
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialog (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Change field contents of the column "text"



Activate dialog (MOC)

Activate dialog for the specific terminal group

9.7.7 Activate simplified dialogs

There are simplified dialogs for: logging on operations, partial confirmations/uploads for operations, interrupting operations, logging off operations.

In order to use simplified dialogs, they may be defined for the default dialogs AIPDEF 0 in the workflow via the "enable simple dialogs" button.

Procedure:

- Menu: System administration --> Terminals --> Dynamic dialogs --> Button "Enable simple dialogs"
- Enable dialogs for AIPDEF 0

Only one dialog is entered in the workflow if simple dialogs are in use.

Once simplified dialogs have been activated, it **cannot** be undone by way of configuration. This can only be changed by way of customizing the system, which has to be ordered from MPDV.

Simple dialogs can only be enabled for the HYDRA default dialogs AIPDEF 0.

9.8 Customizing files

9.8.1 Terminal script files

File names and directories of default/customized terminal scripts must be named as follows:

	AIP	Description
MPDV	**.\ctnet\win\ctaip\etc\aip_mpdv.zip**	MPDV Standard (not used)
MPDV	**.\ctnet\win\ctaip\etc\mpdv-aip.zip**	MPDV Standard
CUST	**.\custom\userexit\aip_<customer no.>.zip**	Customization with customer number
CUST	**.\custom\userexit\aip_<project>.zip**	Customization with project abbreviation

“aip_” is added as prefix to terminal script files for the AIP:

	PRIO	AIP	Description
MPDV	**1**	**.\aip_system_mpdv.scr** **.\aip_<dialog>_mpdv.scr**	MPDV Standard (not used)
MPDV	**2**	**.\aip_mpdv-system.scr** **.\aip_mpdv-<dialog>.scr**	MPDV Standard
CUST	**1**	**.\aip_system_<customer no.>.zip** **.\aip_<dialog>_<customer no.>.zip**	Customization with customer number
CUST	**2**	**.\aip_system_<project>.zip** **.\aip_<dialog>_<project>.zip**	Customization with project abbreviation

ZIP files are only unpacked in live operation, once they have been successfully downloaded from the server. In the DEMO mode a ZIP file to be loaded is unpacked if it exists locally!